# Chapter 7

# Ischemic heart disease

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# Pathophysiology

## Key facts

- Stenotic coronary artery disease (CAD) is narrowing of the coronary arteries caused by thickening and loss of elasticity of the arterial walls (atherosclerosis). The pathological changes are described on 📖 pp72–73.
- Stable plaques are mostly responsible for the lesions seen on angiography and stable angina: MI is due to unstable plaque rupture and thrombosis: 50% of MIs occur distal to angiographically 'normal' vessels.

## Morphology

In patients with symptoms that warrant coronary angiography:

- 40% have three-vessel disease (3VD), 30% two-vessel disease (2VD).
- 10–20% have a significant stenosis of the left main stem (LMS).
- 95% with one completely occluded vessel have a significant stenosis in at least one other vessel.
- The ostia of large arteries and sites of branching are usually involved.
- Diffuse distal disease is less common.
- Occasionally a segment is intramyocardial: on angiography these segments are straight, free of branches, and usually free of disease.

## Structural and functional changes

Coronary artery stenoses progress with time, in severity and distribution. Rate of progression is variable, and regression of lesions has been observed. Over a 2-year period, 50% of patients develop new significant lesions. Stenoses of 50% of cross-sectional area limit coronary flow reserve (the increase in blood flow that occurs to match increased myocardial oxygen demand 📖 p22). Coronary blood flow at rest is reduced by stenoses of 90%. At its most severe CAD results in vessel occlusion.

***Left ventricular dysfunction:*** LV function may be abnormal in systole or diastole, during exercise or at rest. In normal people, global LV systolic function improves with exercise. In patients with CAD, areas of localized regional wall motion appear during exercise. These areas can be demonstrated as myocardial perfusion defects. If CAD is severe enough, these areas become confluent and global systolic LV function falls, sometimes so profoundly that exercise causes hypotension. This is reflected by EKG changes. Early diastolic function, which is energy dependent, is eventually impaired: filling times increase as does LVEDP. Dysfunction at rest is a result of irreversible scarring or hibernation (📖 p28, pp74–75).

## Symptoms

***Exertional angina:*** this is common. It results from reduction of coronary flow reserve, and the severity depends on the mismatch between myocardial oxygen supply and demand, i.e., both the severity of the CAD and the amount of work the myocardium is required to do. It is graded according to severity (Box 7.1) (see also 📖 p2 for more details).

***Dyspnea:*** graded in the same way as angina (see Box 7.1). A number of mechanisms contribute. Acutely, transient systolic, and/or diastolic LV dysfunction results from worsening myocardial supply:demand mismatch

caused by increases in preload, afterload, hypotension, or exertion. LV dysfunction leads to ↑LVEDP, causing sudden elevation in pulmonary venous pressure, detected by pulmonary venous baroreceptors which result in the sensation of dyspnea. Orthopnea results from the sudden increase in preload on lying flat. Paroxysmal nocturnal dyspnea is due to pulmonary edema, and poorly understood changes in respiratory drive.

***Nausea:*** parasympathetic stimulation results in nausea and vomiting.

### Box 7.1 New York Heart Association (NYHA) class

- Class I: asymptomatic.
- Class II: symptoms causing slight limitation of ordinary activity.
- Class III: symptoms causing marked limitation of ordinary activity.
- Class IV: symptoms occurring with mild activity, or briefly at rest.

Angina lasting >15min at rest is 'unstable angina'.

***Unstable angina:*** this term, which covers a multitude of syndromes all of which reflect an adverse prognostic turn, applies to patients with severe and persisting angina. It is most commonly caused by non-occlusive thrombus at the site of an unstable plaque. Unstable angina normally recurs as either another episode of angina or as an acute MI. Patients can be divided into low-, medium-, and high-risk categories depending on the duration of their angina, severity of symptoms, presence of EKG changes, and hemodynamic compromise.

### Acute coronary syndrome (ACS)

- This includes unstable angina, NSTEMI, and STEMI.
- Unstable angina is differentiated from NSTEMI by raised cardiac enzymes (CK-MB or troponin) that are present in NSTEMI.
- STEMI is differentiated from NSTEMI by EKG changes.

***Non-ST segment elevation MI (NSTEMI):*** this is closely related to unstable angina. The key difference is severity, characterized by the presence of CK-MB and troponin I and T, in NSTEMI. Troponin correlates to short-term mortality in patients with ACS: in one study <2% of patients with troponin I levels of 0.4–1.0ng/mL were dead 42 days after presentation, compared to 7.5% with troponin I levels >9.0ng/mL.

***ST segment elevation MI (STEMI):*** this is caused by acute total or subtotal vessel thrombotic occlusion (📖 p72). Patients with severe proximal LAD lesions are particularly at risk of large MIs. In patients with symptoms severe enough to warrant coronary angiography 10% have an acute MI within 1 year and 30% have an acute MI within 5 years. Hospital mortality of STEMI is 7–10%, and can be predicted by size of infarct, reserve in the adjacent viable myocardium, and the number of prior MIs.

***Death:*** overall survival for a group of patients with clinically evident CAD is 75% at 5 years after starting medical treatment.

- In 1VD the survival is the same as normal (unless proximal LAD).
- For 2VD the 5-year survival is 75%, lower if LV function is impaired.
- In 3VD the 5-year survival is 50%, lower if LV function is impaired.
- Tight, symptomatic LMS disease has a 2-year survival of 50%.

# The evidence for CABG

## Key facts

CABG has been compared to medical therapy, percutaneous transluminal coronary angioplasty (PTCA), and stenting (bare metal and drug-eluting), in randomized controlled trials (RCTs) and large registries.

## Key studies

- *CABG vs. medical therapy*: 3 large RCTs—Veterans Administration (VA), European Coronary Surgery Study Group, and Coronary Artery Surgery Study (CASS); 4 smaller ones (Texas, Oregon, and 2 New Zealand studies), and several large registries.
- *CABG vs. PTCA*: 9 large RCTs: BARI, EAST, GABI, Toulouse, RITA, ERACI, MASS, Lausanne, CABRI.
- *CABG vs. PCI*: 15 large RCTs: SYNTAX, ARTS, SoS, AWESOME, ERACI II, SIMA, Diegler. Several large registries, e.g., Hannan et al.
- *PCI vs. medical therapy*: large RCTs, e.g., COURAGE, and meta-analyses.

## Problems with the data

- RCTs and their meta-analyses are the 'highest' form of evidence as biases inherent in registries are eliminated, but problems include:
  - Patient selection in most early studies excluded individuals >65 years, women, and high-risk patients such as those with multi-vessel or left main disease, resulting in study populations that were not representative of clinical practice, and dominated by patients least likely to benefit from CABG, e.g., 60% of CASS patients had 2VD.
  - Medical and surgical treatment have changed since early RCTs.
  - Large numbers of patients crossed over to the intervention arms.
  - Limited follow-up (<5 years) under-represents benefits of CABG.
  - Combined endpoints give equal weight to death and reintervention.
- Registries may better reflect 'real-world' practice, but:
  - Non-randomized data cannot account for all confounders influencing both choice and outcome of a treatment strategy so two different patient populations are being compared, not just two treatments.

## CABG vs. medical therapy

These trials form the basis of later studies but are of largely historical interest as only one trial used arterial grafts (CASS), and in only 14% of patients, newer methods of cardioprotection were not used, ASA, statins and ACE inhibitors were not standard, β-blockers were used in <50% of patients, and up to 50% of medical patients crossed over to surgery:

- Median survival for LMS was 13.3 years with CABG vs. 6.6 years without.
- Median survival for proximal LAD disease was 13.1 years with CABG vs. 6.2 years with medical therapy.
- Patients with worse LV gained most survival benefit from surgery.
- There was no difference in MI, and survival converged after 12 years because of SVG occlusion and medical patients crossing over to surgery.

## CABG vs. PTCA

These studies had similar flaws to CABG vs. medical studies. The primary problem was that patients in whom survival had already been shown to be better with CABG than medical therapy were not included (only 5% of patients with multivessel disease at participating centres were enrolled: half did not meet the inclusion criteria (which excluded LMS disease), and most that did were not randomized—so the study populations were biased to PTCA and unrepresentative of clinical practice:

- 5-year survival was similar for CABG (89.3%) and PTCA (86.3%).
- Diabetics had much better survival with CABG than PTCA.
- By 1 year 20% of patients randomized to PTCA required CABG.
- At 5 years the reintervention rate was 7× higher with PTCA.

## CABG vs. PCI

***SYNTAX***

- 1800 'all-comers' including LMS and multivessel disease randomized plus registry of 1300 ineligible patients, planned 5-year follow-up.
- Death at 3 years double in PCI patients (11% vs 4.5% in CABG patients, p=0.03).
- In the tercile of patients with the lowest SYNTAX (coronary complexity) scores death and MACCE was the same for PCI and CABG.

***The Arterial Revascularization Therapies Study Group (ARTS) trial***

- 1205 patients with multivessel disease.
- No difference was detected at 1 year in MACCE.
- There was a higher rate of repeat revascularization in the stenting group (16.8%) compared to the CABG group (3.5%).

***Stent or Surgery (SoS) trial***

- 988 patients with multivessel disease.
- There was a higher rate of repeat revascularization in the stenting group (21%) compared to the CABG group (6%).
- 3× more deaths in PCI group (CABG 0.8% 30-day mortality).
- *No* difference in cognitive function at 1 year.

***Coronary angioplasty with stenting vs. coronary artery bypass in patients with multiple-vessel disease (ERACI II) trial***

- 450 patients were randomized to stent or CABG. 40% 2VD.
- This trial is the only one to show an increased risk of death at 1 year in the CABG group (8%) vs. the stent group (3%), but the annual volume at participating centres only averaged <60 CABG total.
- Repeat revascularization was higher in the stent group (17%) compared to the CABG group (5%).
- There was no difference between the groups for MACCE.

## PCI vs. optimal medical therapy

COURAGE randomized 2287 patients with significant CAD and ischemia: at median follow-up of 4.6 years there was no difference in MACCE (death, MI, stroke, hospitalization for unstable angina). 30% of medical patients vs. 20% of PCI needed revascularization. Only 30% were 3VD, and LMS was excluded. Most meta-analyses report similar results.

# Indications for CABG

## Key facts

Surgery is for prognosis or symptoms. The indications are adapted from consensus guidelines published in 2009 and 2010 (references on 📖 p825).

## Classification system for recommendations

- Class I: conditions for which there is evidence and/or general agreement that a given procedure is useful and effective.
- Class II: conditions about which there is conflicting evidence:
- II a: weight of evidence/opinion is in favour of usefulness/efficacy.
- II b: usefulness/efficacy is less well established by evidence/opinion.
- Class III: conditions for which there is evidence and/or general agreement that a given procedure is not effective and may be harmful.
- **Level of evidence:** A, multiple RCTs or meta-analyses; B, single RCT or non-randomized studies; C, consensus, small/retrospective studies.

## Stable angina

See Table 7.1.

**Table 7.1** Indications for revascularization in patients with lesions suitable for both and low predicted surgical mortality

| Revascularization | Subset of CAD | Class | Level |
|---|---|---|---|
| **For prognosis** | Left main >50% | I | A |
| | Proximal LAD >50% | I | A |
| | 2VD or 3VD with ↓LVEF | I | B |
| | Large area ischemia (>10% LV) | I | B |
| | 1VD without proximal LAD and without >10% ischemia | III | A |
| **For symptoms** | Any stenosis >50% with limiting angina or equivalent on OMTx | I | A |
| | Dyspnea/CHF and >10% LV ischemia supplied by vessel with >50% stenosis | IIa | B |
| | No limiting symptoms with OMTx | III | C |
| **Indications for CABG vs. PCI** | | **CABG** | **PCI** |
| 1VD or 2VD—non-proximal LAD | | IIb C | I C |
| 1VD or 2 VD—proximal LAD | | I A | IIa B |
| 3VD simple disease, full functional revascularization achievable with PCI, SYNTAX score <22 | | I A | IIa B |
| 3VD complex lesions, incomplete revascularization achievable with PCI, SYNTAX score >22 | | I A | III A |
| LMS (isolated or 1VD, ostium/shaft) | | I A | IIa B |
| LMS (isolated or 1VD, distal bifurcation) | | I A | IIb B |
| LMS + 2VD or 3VD, SYNTAX score <32 | | I A | IIb B |
| LMS + 2VD or 3VD, SYNTAX score >33 | | I A | III B |

## Acute coronary syndromes

### *NSTEMI*

Mortality is high, and similar to that for STEMI patients after the first month. NSTEMI patients are a heterogeneous group. Aims of revascularization are: (1) symptom relief and (2) prognosis. RCTS show early invasive strategy reduces severe recurrent ischemia, and need for hospitalization and reintervention, with reduction in mortality and MI in medium term.

- Choice of PCI or CABG depends on the severity and distribution of CAD (similar to Table 7.1), and predicted risk of surgery.
- The optimal timing of surgery and PCI is different: early PCI provides greatest benefit in ACS, but benefit from CABG is greatest when patients can undergo surgery after several days of medical stabilization.

### *STEMI*

Primary PCI in setting of STEMI in experienced centres within 6–12h after symptom onset provides better angiographic and clinical outcomes than fibrinolysis, which remains an important alternative because of variability in access to PCI. There is a limited role for CABG in acute STEMI:

- In cases of unfavorable PCI anatomy or PCI failure, emergency CABG involving STEMI should only be considered when a very large myocardial area is in jeopardy and surgical revascularization can be completed before this area becomes necrotic (3–4h).
- Like NSTEMI—outcomes with CABG improve the more time elapses between the acute event and surgery. In the absence of persistent pain or hemodynamic compromise waiting 3–7 days is recommended.
- Surgery is indicated but associated with very high mortality for mechanical complications of STEMI including free wall rupture, ischemic VSD and papillary muscle rupture.

## Heart failure and CABG

The indications for revascularization for CAD in the setting of CHF are similar to those outlined in Table 7.1, but the operative mortality is increased which impacts on aspects of the decision to operate:

- Patients without demonstrated viability are unlikely to benefit from revascularization and high-risk surgery should therefore be avoided.
- Patients with severely dilated LV have a low likelihood of showing improved LVEF even the presence of substantial viability (📖 p28).
- MRI is the standard imaging technique to assess myocardial anatomy, regional and global function, viability, infarct size, and transmurality.
- In patients with LVEF <35% and predominantly anginal symptoms, CABG is recommended over PCI for significant LMS, LMS equivalent, and proximal LAD disease with 2VD or 3VD.
- In patients with LVEF <35% and predominantly heart failure symptoms (no or mild angina) CABG can be considered if there is viability, with aneurysmectomy if there is a large LV aneurysm (📖 p348).

## Redo CABG

PCI is mainstay of therapy, but redo CABG may be indicated for patients with several diseased grafts, reduced LV function, several chronic total occlusions in absence of patent LIMA to LAD graft, or severe symptoms/ischemia despite optimal medical therapy (📖 pp342–347).

# When not to operate

## Key facts

- ACC/AHA guidelines identify several scenarios where there is evidence that CABG is likely not to be efficacious, or likely to do harm.
- If you remember that you are operating for symptoms and/or prognosis, then lack of a benefit in one area places a great deal of scrutiny on your rationale for operating for the other.

## Anatomy

CABG is not indicated in borderline coronary stenoses (50–60%) diameter in locations other than the LMS without demonstrable ischemia on non-invasive testing, or for insignificant (<50% diameter) stenoses.

## Asymptomatic patients

The aim of surgery in these patients is to improve prognosis. Surgery is only beneficial in asymptomatic patients at high risk of MI or death (📖 p296) because of severity of lesions, i.e., in severe LMS disease, LMS equivalent, or proximal LAD disease.

- Take a careful history: ischemia may present as epigastric discomfort; arm, neck, or jaw pain; or dyspnea rather than angina.
- Beware of the patient with dyspnea as their predominant symptom rather than angina. Documenting the presence of ischemia by exercise EKG, echocardiography, or thallium scan (📖 p28), carefully assess ventricular dysfunction (systolic and diastolic) and take care to rule out non-cardiac causes of their dyspnea.
- In very elderly patients, or patients with severe comorbidity, who are unlikely to see a survival benefit from surgical revascularization, lack of symptoms is a strong argument for not operating—you risk decreasing quality of life, without an incremental increase in length of life. This scenario sometimes arises when CAD is identified during the work-up for another procedure, e.g., hip replacement. Think carefully: PCI may be the most reasonable approach.

## Patients with chronic stable angina

Patients with 1VD or 2VD not involving proximal LAD stenosis who (1) have mild symptoms that are not necessarily due to myocardial ischemia or, (2) have not received an adequate trial of medical therapy, and have only a small area of viable myocardium or no demonstrated ischemia on non-invasive testing should not be offered surgery, as it is unlikely to have any prognostic or symptomatic benefit.

## Patients with unstable angina/NSTEMI

Unstable angina is best managed initially by medical treatment. The risk of CABG, which is performed for prognostic and symptomatic indications described in Table 7.1, decreases with distance between the acute event and surgery: waiting 48–72h if the patient has no ongoing signs of ischemia, on iv heparin, NTG, and an IABP if necessary, is the best strategy.

### Patients with acute MI

Primary reperfusion >12h after the start of an evolving ST-segment MI without evidence of ongoing ischemia is high risk and has no survival benefit. In the patient with cardiogenic shock post acute MI, the most reasonable surgical strategy may be ECMO or a temporary VAD (📖 pp496–501) rather than CABG.

### Patients with poor LV and no ischemia

Operative mortality for isolated CABG in selected patients with chronic severe LV dysfunction is around 5%, compared to patients with preserved LV function where it is 1–2%. Long-term survival is poorer. However, selected patients with severe LV dysfunction and CAD still derive significantly more benefit from surgical revascularization than they do from PCI or medical management, in terms of symptoms, freedom from revascularization and survival. Patients with poor LV being considered for CABG should have: (1) *severe, obstructive CAD* with (2) *adequate targets*, *and* documented evidence of (3) *ischemia* and (4) *viability*.

- If dyspnea and other failure symptoms predominate over angina then the benefits of CABG are less clear: it is a relatively high-risk strategy compared to PCI or medical management (📖 p493), with debateable prognostic or symptomatic benefit.
- Patients with severely dilated LV have low likelihood of showing improvement in LVEF, even if they have large areas of viability: concomitant ventricular restoration (📖 p495) may have a role in ameliorating symptoms of heart failure, or in end-stage ischemic cardiomyopathy assist devices (📖 p496) and transplantation (📖 p504).

### Patients post failed PCI

CABG is not indicated if there is no evidence of ischemia, or if PCI was abandoned simply because of target anatomy. The absence of reflow after PCI is not an indication for CABG.

- If you are called to cath lab to be confronted by a patient who has been undergoing chest compressions for more than just a few minutes, first establish whether the primary issue is tamponade (surgery indicated if pericardiocentesis not possible) or ischemia, and what the patient's baseline comorbidity and functional status was.
- Emergency CABG has high mortality in this situation: it is often more reasonable to place the patient on ECMO or a temporary assist device (📖 pp500–501) rather than performing CABG.

### Elderly patients

Age per se is not a contraindication to surgery. The normal life expectancy of an octogenarian is about 6 years. Look carefully though for factors such as porcelain aorta, end-organ dysfunction that might push the balance in favour of PCI.

### Small or occluded distal vessels

Angiographic appearances may be deceptive (vessels with tight proximal stenoses may just be underfilled), and in the majority of cases it is possible to place a graft at operation.

# Planning coronary artery surgery

## Key facts

There are different ways of performing almost every element of a CABG. Several factors are taken into account when planning a CABG to provide minimum morbidity and mortality, and longest graft patency and life expectancy.

## History

Document:

- Age, frailty (extreme age/frailty may favour PCI, or fewer grafts).
- Severity and nature of symptoms: anginal type symptoms usually predominate, whereas patients presenting with predominantly heart failure type symptoms (e.g., dyspnea) may need further evaluation.
- Number and timing of previous MIs impact risk and timing of surgery.
- Coronary anatomy, LV function, valvular heart disease.
- Previous PCI and operations.
- Risk factors for perioperative mortality including renal dysfunction, pulmonary dysfunction, peripheral vascular disease, strokes.
- Risk factors for wound infections including smoking history, diabetes, steroid use, significant involuntary weight loss, malnutrition.
- Saphenous vein stripping, DVT, chest wall radiation, dominant hand.

## Examination

- A targeted clinical examination is carried out to screen for:
- Obesity, pectus deformities, previous surgery.
- Presence of other cardiovascular problems such as poorly controlled hypertension, arrhythmias, valvular heart disease, carotid artery disease, peripheral vascular disease, evidence of heart failure.
- Varicose veins, venous ulceration.
- Allen's test (📖 p7).
- Respiratory pathology.

## Blood tests

- Standard screening tests are described on 📖 pp8–9. A FBC, clotting, serum electrolytes, LFTs, and group and save are routinely performed.

## CXR

A PA CXR is used to assess:

- Pulmonary pathology.
- Evidence of heart failure.
- Calcification of the ascending aorta.
- Method of closure, if previous sternotomy.
- Rarely mediastinal masses.

## Cardiac catheterization

***Coronary angiography***

Coronary angiogram should not be >6 months old. The objective of CABG is to achieve complete revascularization.

- All stenoses >50% in diameter in all coronary artery vessels should be bypassed: ideally there should be good distal run-off.
- If the right coronary artery is non-dominant it is not grafted.
- Similarly, small, short branches in the left coronary circulation are not always grafted.
- Some surgeons elect not to bypass branches that are very close to arteries that will be revascularized, arguing that the myocardium will be adequately supplied by nearby arteries.
- Most surgeons will only bypass the more significant vessel of two vessels that communicate distal to a stenosis.
- A significant LMS lesion means that all major coronary arteries in the left system need revascularization.
- If there is a distal stenosis in addition to a proximal LAD or LMS stenosis it may necessary to either bridge the distal lesion so that the whole length of the LAD is supplied, or revascularize both segments with either a jump graft or two separate grafts.
- Lesions in the mid portion of important vessels are sometimes bridged, to provide supply proximally and distally.

***Cardiac catheterization***

- Ventriculography is usually performed in the absence of renal dysfunction or haemodynamic instability giving an estimate of LV function and regional wall motion abnormalities, and MR (📖 p22).
- Gradient across the aortic valve can be assessed by measuring LV and aortic pressure either simultaneously or by pull-back.
- Right heart catheterization should be performed in patients with very poor LV function to quantify pulmonary hypertension.

## Echocardiography

All patients undergoing coronary artery bypass surgery should undergo preoperative TTE to identify valvular heart disease (e.g., ischemic MR) and accurately quantify LV function.

## Myocardial perfusion studies

The principles of myocardial perfusion studies, which include MRI, technetium scans, stress echocardiography, and PET scans, are described on 📖 p28. These are requested to determine whether myocardium is akinetic as a result of irreversible scarring from myocardial infarction (in which case there is no benefit from revascularizing the vessel supplying that territory), or as a result of hibernation or stunning (in which case there is a benefit to be gained from revascularization) (📖 p297).

## Carotid artery duplex

These are performed in patients that give a previous history of stroke or TIA, have carotid bruits, or have evidence of severe polyarteriopathy.

## CT scan

Increasingly routinely performed in patients over the age of 75 years to identify aortic calcification that may change intraoperative strategy, and occult pulmonary and mediastinal pathology.

# Conduit selection

## Key facts

Choice of conduit is determined by the quality of the available conduit, the suitability of the targets, and the age and comorbidity of the patient (Table 7.2).

**Table 7.2** Approximate conduit patency rates

| Conduit | 1-year patency | Late patency rates |
|---|---|---|
| Left internal mammary artery | 98% | 90% at 10 years |
| Right internal mammary artery | 96% to LAD, Cx 80–90% to RCA | 90% at 5 years |
| Radial artery | Highly dependent on target coronary | |
| Right gastroepiploic artery | 95% | 90% at 3 years |
| Right inferior epigastric artery | 80% | Not known |
| Long saphenous vein | 80–90% | 50% at 10 years |
| Short saphenous vein | Not known but probably similar to LSV | |
| Cephalic vein | <60% | |
| Saphenous vein (stored) | 40% | |

## Internal thoracic arteries

**LIMA to LAD graft:** this is the mainstay of CABG as it provides better survival and freedom from MI, symptoms and PCI than SVG to LAD.[1] Over 90% of LIMA to LAD anastomoses are patent 10–20 years postoperatively, and late graft occlusion is uncommon. Stenoses, which occur in 5–10% of LIMA to LAD anastomoses, do not usually progress to occlusion. Free LIMA to LAD graft has a patency rate slighter lower than pedicled LIMA. The excellent patency of mammary artery grafts is probably due to endothelial cell function, which seems to help resist atherosclerosis, and grafting to the LAD or Cx territory which has a larger run-off than the RCA territory.

**BIMA vs. LIMA** (see also 📖 p321): bilateral internal mammary artery (BIMA) shows better long-term patency than LIMA + SVG and may improve symptom-free survival (this is difficult to judge from retrospective studies which do not adjust for all the confounders that affect choice of conduit), but most studies, including recent randomized data, show double the risk of deep sternal wound infection. For that reason most surgeons avoid BIMA in obese, diabetic, current smokers, or older patients. The in-situ RIMA will only reach proximal right-sided or very proximal left-sided lesions. 'T' grafting free RIMA off the LIMA allows the RIMA to be grafted to the more distal Cx and PDA sequentially (📖 p321). The in-situ RIMA may be anastomosed to the LAD, and the LIMA to the Cx system. Alternatively, the in-situ RIMA may be passed through the transverse sinus to the Cx system, where bleeding or torsion may be difficult to address, but this minimizes problems posed by RIMA crossing the midline at future sternal re-entry.

***Skeletonized vs. pedicled IMA:*** skeletonizing the IMA (📖 p320) facilitates sequential and distal grafting, and may help reduce deep sternal wound

infection particularly where BIMA is used. Patency is comparable to pedicled LIMA at 10 years.

***Relative contraindications to LIMA***

- Previous radiotherapy to the left chest.
- Atherosclerotic or stenosed left subclavian artery (could use free graft).
- Emergency surgery for cardiogenic shock.



***Relative contraindications to BIMA***

- Diabetes, particularly females.
- Obese patients, particularly females.
- Current smokers, COPD, or other respiratory impairment.
- Immunocompromised patients, long-term high-dose systemic steroids.

## Radial artery grafts

Patency rates of radial artery grafts (📖 p316) are dependent on the distal vessel: patency rates exceed 90% at 1 year and 80% at 5 years, but are much lower if the stenosis being bypassed is <70%, has poor run-off, or is a right-sided vessel—in these situations radial patency has been shown to be lower than SVG. Occluded or thread-like arterial grafts are found where the stenosis being bypassed is <70%. The radial is much more muscular and prone to spasm than the IMA.

## Saphenous vein

SVG has an early patency rate of 80–90% and a 50% patency rate at 10 years. Late patency is about 80% when it is used to bypass the LAD which has a large run-off. Patency may be reduced by excessive manipulation during endoscopic harvest. Stenosis is caused by:

***Thrombosis:*** this is the commonest cause of early graft failure, and is usually due to technical factors. Endothelial cell loss and exposure of the basement membrane predisposes to graft thrombosis by causing accumulation of platelets, fibrin, and thrombus on the luminal surface. Thrombosis also occurs in heavily atherosclerotic areas. When antiplatelet therapy is not used up to 10% of grafts occlude within a few weeks of surgery.

***Anastomotic narrowing:*** this occurs proximally in about 20% of vein grafts at 1 year, and distally in about 50% of patients at 1 year, and may be due to technical factors, atherosclerosis, or intimal hyperplasia.

***Intimal hyperplasia:*** this diffuse thickening of the intima, occurs almost universally in vein grafts, and appears to be a remodelling process that approximates the vein lumen to that of the distal arterial lumen. This diffuse narrowing is detectable in most SVGs angiographically at 1 year.

***Fibrous atherosclerotic plaques:*** these begin to develop in vein grafts 3 years postoperatively and are present in most vein grafts by 10 years. They are responsible for flow-limiting stenoses. They are related to damage to the graft endothelium as a result of harvest and insertion into a high-pressure arterial system, and the patient's tendency to develop atherosclerosis. Aggressive cholesterol management can slow this process. Vein graft atherosclerosis differs from native vessel disease: it is diffuse, and plaques are superficial, friable with overlying mural thrombus.

## Reference

1. Loop *et al.* Influence of the IMA graft on 10 year survival and other cardiac events. *NEJM* 1986; **314**:1–6.

# Off-pump versus on-pump CABG

## Key facts

CABG can be carried out with cardiopulmonary bypass (on-pump or ONCAB), or without (off-pump or OPCAB). Most surgeons perform CABG on-pump: the advantages of a still, clear operating field, myocardial protection, and hemodynamic and respiratory control, outweigh many of the disadvantages of bypass (📖 p106). Improvements in methods of stabilizing the heart have led to some surgeons adopting OPCAB to avoid the complications of cannulation and cardiopulmonary bypass.

## The evidence

### *Randomized controlled trials*

Over 50 RCTs of off-pump versus on-pump surgery have been published, mostly underpowered to detect a difference in stroke or mortality, with some showing reductions in postoperative atrial fibrillation, transfusion requirements, inotrope requirements, ventilation times, length of hospital and intensive care unit stay, and cost compared to on-pump coronary artery surgery. Some of the problems outlined on 📖 p294 apply to these studies: additionally the off-pump learning curve may impact trial results from less experienced trial centers.

- In the largest RCT reported to date, the ROOBY (Randomized On/Off Bypass) investigators randomized 2203 patients undergoing urgent or elective CABG. Unlike the majority of off-pump trials, poor LV, advanced age and acute coronary syndromes were not exclusion criteria. Vessels <1.1mm or diffuse disease were exclusion criteria.
- At 1 year the composite outcome of death, MI, or any revascularization after 30 days, was significantly higher in the off-pump group, as was death from cardiac causes. Patency of both saphenous and LIMA grafts was lower in the off-pump group. No difference was seen in renal failure, reop for bleeding, stroke, or cognitive function.
- No RCTs have shown a difference in 30-day mortality or stroke: Van Dijk et al., Puskas et al., and Angelini et al. did, however, show a significant reduction in transfusion requirements, serum markers of myocardial damage, and a cost saving with off-pump surgery in low-risk patients in trials that randomized 281, 197, and 401 (2 pooled randomized controlled trials) patients respectively.

### *Meta-analyses*

A large, recent meta-analysis including 86 trials containing 10716 patients found increased mortality with off-pump CABG (3.7% vs. 3.1%, p=0.04), and fewer distal anastomoses with off-pump. No differences in MI or stroke was seen.

### *Registries*

Several retrospective analyses of large registries, on the other hand, have demonstrated a statistically significant difference in postoperative mortality and stroke. Differences in outcome may be due to patient selection as it isn't possible to control for all confounding variables in retrospective studies, even with risk adjustment.

- Hannan *et al.* compared over 13 000 off-pump and 35 941 on-pump patients operated on between 2001–2004.
- The off-pump group had lower risk adjusted mortality fewer strokes less respiratory failure postoperatively but there was no difference in 3 year mortality.

## Summary

***Mortality:*** recent data from a large RCT and meta-analysis shows increased operative mortality with OPCAB compared to ONCAB.

***Stroke:*** no RCTs have looked at 'no-touch' technique versus CPB. Randomized trials have not shown a difference in stroke rates. The majority of retrospective analyses have shown reduced rates of stroke in off-pump (2%) versus 1% in on-pump CAB: may be due to patient selection.

***Postoperative MI:*** no difference in the incidence of postoperative MI has been shown, but fewer cardiac enzymes are released in OPCAB.

***Incomplete revascularization:*** ROOBY trial found significantly fewer grafts performed than planned in the off-pump group.

***Graft patency rates:*** ROOBY trial found a third higher incidence of at least 1 occluded graft at 1 year (37% vs. 29%), reduced SVG patency (77% vs 88%), and fewer excellent LIMA grafts (89% vs 93%).[1]

***Length of stay:*** length of hospital and ICU stay depend on several non-patient related variables. ROOBY showed no difference in length of ICU/hospital stay, or time on a ventilator.

***Renal failure:*** retrospective studies have shown reduced rates of renal failure in OBCAP patients, but not seen in ROOBY.

***Bleeding, transfusion, and resternotomy:*** these have been shown to be reduced in OPCAB, in some but not all randomized and retrospective studies. No difference in any of these was seen in ROOBY trial.

***Atrial fibrillation:*** the prevalence of postoperative AF in OPCAB patients is lower according to some retrospective and randomized studies.

## Current indications for OPCAB surgery

- OPCAB is a suitable alternative to CPB in experienced centers. Combined with a 'no touch' or anaortic technique (📖 p321) it is the first approach for patients with a very atherosclerotic or calcified aorta, and may be associated with much reduced risk of stroke in a broader patient population.

## Reference

1. Shroyer AL, Grover FL, Hattler B *et al.* On-pump versus off-pump coronary-artery bypass surgery. *N Engl J Med*. 2009;**361**:1827–37.
2. Moller CH, Penninga L, Wetterslev J *et al.* Off-pump versus on-pump coronary artery bypass grafting for ischaemic heart disease. *Cochrane Database Syst Rev*. 2012;3:CD007224.
3. Hannan EL, Wu C, Smith CR *et al.* Off-pump versus on-pump coronary artery bypass graft surgery. *Circulation*. 2007;116:1145–52.

# Open saphenous vein harvest 1

The anatomy of the long saphenous vein is described on 📖 p55. Assessing the conduit is described on 📖 p7, and complications of saphenous vein harvest are described on 📖 pp283 and 285.

## Indications

- Suitable for the majority of patients undergoing CABG.
- Occasionally a segment is required unexpectedly if a coronary vessel is inadvertently damaged during non-CABG surgery, or for salvage grafting if the patient fails to come off bypass.
- Short segment may be used in patch repair in carotid endarterectomy.

## Contraindications

- Overlying sepsis.
- Multiple large varicosities in the LSV itself.
- Deep venous insufficiency in that leg.
- Previous DVT in that leg.
- Inadequate diameter lumen.

## Technique

The aim is to obtain the required length of conduit and close the wound before the heparin is given to minimize bypass time and hematomas.

### Options for saphenous vein harvest

- Open technique starting at ankle.
- Open technique starting at groin.
- Closed technique using vein stripper.
- Endoscopic harvest.
- Short saphenous vein harvest.

*Prepping and draping*

The legs and groin should have been shaved in the anesthetic room. Shaving on the ward the night before surgery results in increased rates of wound infection. The legs are prepped after the trunk. Prep the area of the incision first, then from clean areas to dirty. Leave the groin until last. An unscrubbed assistant lifts the legs so that you can prep the undersurface. If a diathermy plate is on the buttocks, take care not to use so much prep that it dribbles down onto the plate as this can result in diathermy burns. Finally ask the assistant to spread the legs so that you can prep the groin. Lower the prepped legs so that the feet can be draped, making sure the medial malleolus is in the operative field.

*Surface anatomy*

### Surface anatomy of the long saphenous vein

The long saphenous vein runs between the following points (Fig. 7.1a):

- 1cm above and anterior to the medial malleolus.
- 4cm posterior to the medial border of the patella.
- 2cm below and 1cm medial to the mid-inguinal point.

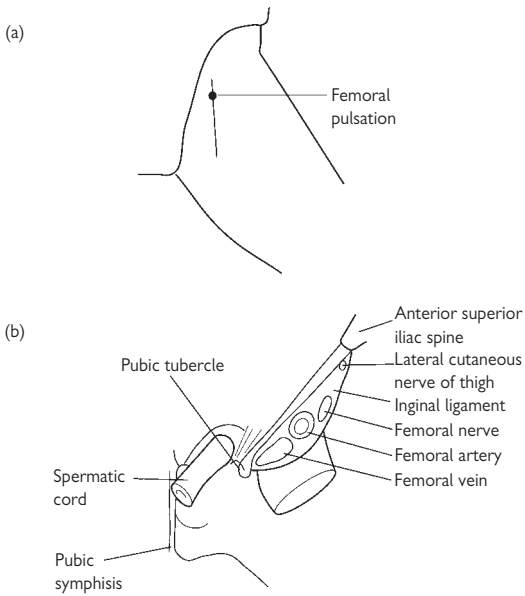



**Fig. 7.1** (a) Incision for long saphenous vein harvest in the groin. (b) Instruments on the saphenous vein set: (i) McIndoe scissors, (ii) Metzenbaum scissors, (iii) Mayo scissors, (iv) DeBakey forceps, (v) Mosquito clips (vi) Dunhill clips, (vii) Ligaclip applicator.

# Open saphenous vein harvest 2

## Key facts

- Your priority is to identify adequate quality conduit and harvest adequate length efficiently.
- First locate the vein, if you have any doubts that the lumen is adequate next dissect out 2–3cm and cannulate it to check that it distends OK ***before*** harvesting a complete length (which it may be worth fully dissecting out so you can hand it out), before going on to harvest more vein.

## Technique

Position the leg so that it is abducted at the hip, and flexed by 45° at the knee. Check how many lengths of vein are needed. A length is about 25cm (or approximately the length of a McIndoe scissors). Grafts to proximal diagonal and obtuse marginal vessels are shorter than grafts to distal posterior descending artery vessels, and are also influenced by the size of the heart.

- If the sternotomy incision has not been started, ask the anesthesiologist if it is OK to start: if the patient is extremely hypertensive, the anesthesiologist may ask you to wait momentarily until they improve BP control.
- Make a 6cm longitudinal incision with a 23 blade, through to fat, starting 1cm above and anterior to the medial malleolus.
- The long saphenous vein should be visible running immediately beneath this: its position is constant.
- If you can't see the vein spread the soft tissue with a McIndoe scissors until it comes into view: if you see muscle/ligament you are too deep.
- Keeping in the plane of the vein, pass one blade of a curved Mayo scissors over the vein, separating the fascia and skin off the vein. Lifting the scissors up, away from the vein, divide the skin cleanly.
- Rapidly follow the vein up the leg in this way so that the required length of vein is exposed.
- Assess the lumen as you go: if it narrows markedly follow it 2–3cm to see if it increases, and check that surgeon is happy with the lumen. Anything <0.5cm or >1cm is probably not useable.
- Your next priority is to ensure that you have a length ready to use.
- The most distal length has fewest branches: using blunt dissection with the McIndoe scissors carefully free the vein from surrounding fascia, so that the side branches are revealed.
- Taking care not to avulse the side branches pass the jaws of a mosquito clip with a 3/0 silk or Vicryl tie underneath the side branch, and put three careful throws flush with the long saphenous vein.
- Ligaclip the leg end of the sidebranch and divide it.
- Proceed up the leg. As soon as one length is free, insert a tester into the most distal end, placing an artery clip across the vein distally.
- Inflate the vein with saline or heparinized blood, compressing it distally between a thumb and forefinger, to see if there are any leaks.

- If there is a leak from an untied side branch place a mosquito clip on the end of the side branch and tie it off flush with the vein.
- If there is a hole in the vein, from a varicosity or avulsed side branch, this needs to be repaired with 6/0 Prolene (Fig. 7.4).
- Dissect out the rest of the vein in the same way, check for leaks, and check that the surgeon is happy with the length, place a Dunhill on the vein proximately, and cut both ends.
- Place a Redivac drain if the patient is very overweight or there is a flap.
- Close the legs with continuous 1/0 Vicryl to deep fascia and 3/0 Vicryl or Monocryl on a curved needle, using subcuticular stitch (Fig. 7.2c).
- Clean the leg, dress the wound with Mepore and bandage the leg firmly, but not too tightly, from ankle to thigh (Fig. 7.2a).

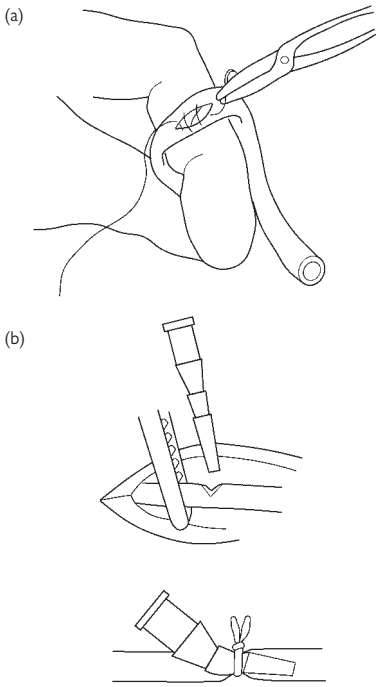



**Fig. 7.2** (a) Repairing the vein. (b) Tying in the tester.

# Open saphenous vein harvest 3

## Key facts

- Harvesting conduit quickly and efficiently is a key skill that is frequently left to the most junior members of the team to perform unsupervised.
- Doing it well means the difference between a pain-free patient, who mobilizes quickly and has an angina-free interval of several years after surgery; and a patient who fails to mobilize because of pain and wound infections, and may experience resternotomy for bleeding within hours or graft occlusion within months.

## Incision

The aim is to make the cleanest incision so that skin edges are not macerated, keeping directly over the vein to avoid creating a flap. The cleanest incision is made with a fresh scalpel, e.g. a 23 blade, but most surgeons compromise, using a Mayo scissors as they find it faster. The options are:

- Make the entire incision with a scalpel. This is quick and clean, but risks creating a flap if the course of the saphenous vein deviates much.
- Make multiple small skin incisions with a scalpel, leaving skin bridges (Fig. 7.3). This is fast and tends to heal better, but there is a risk of avulsing small branches of saphenous vein underneath the skin bridges.
- Cut down using a scalpel onto the Mayo scissors, which is held in the tunnel above the vein. This is safe but time consuming.
- Use the Mayo scissors to cut the skin. This is quick but macerates the skin edges unless the Mayo scissors are adequately sharp.

## Exposure

One of the first things that the more experienced surgeon does when asked to help out a junior colleague in trouble is to adjust the light, place a retractor, and widen the incision. Bad exposure makes an easy operation difficult, and a difficult one almost impossible. Make sure your light is adequate at all times. Use a self-retaining retractor in the thigh and in fat patients. Remember that it is much more difficult to find the vein through a small, ineffective incision, than it is through a large, accurate one.

## Surgical knots

These are a cornerstone of surgery. Loose ligatures can be the cause of a resternotomy for bleeding, and suggest the surgeon is unlikely to progress to tying coronary anastomoses with 8/0 Prolene. There are a variety of ways of tying surgical knots but the principles are the same:

- Put down each throw so that it lies flat (Fig. 7.3).
- One throw followed by the same throw makes a slip-knot (Fig. 7.3c).
- One throw followed by a different throw locks the knot (Fig. 7.3b).
- In braided suture put down 3–5 alternate throws and cut 2–3mm.
- In monofilament, put down 6–8 throws and cut 0.5cm.
- Lay each throw down with your index finger controlling the amount of tension you place on the knot most securely: too much tension can pull the tissue off the wound and too little leads to loose ligatures.

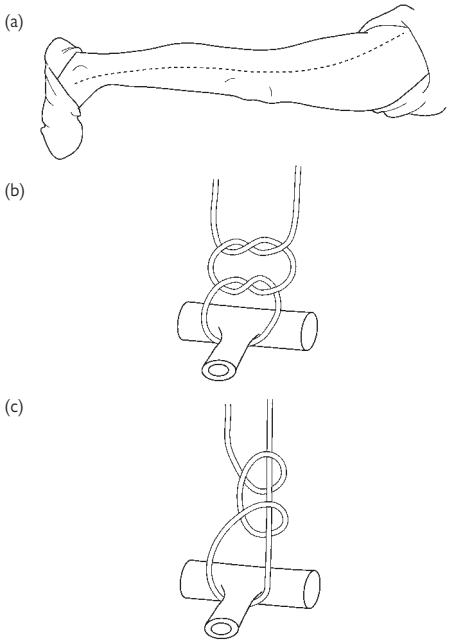



**Fig. 7.3** (a) Saphenous vein incision. (b) Surgical knots: lying throws flat. (c) Slip knot.

# Open saphenous vein harvest 4

## Hemostasis

If you are unsure of your ties, transfix both ends of the saphenous vein stumps before tying them. If you can see a vein end, ligate it even if it is not actively bleeding: when venospasm wears off and BP rises it will start to bleed. There is no point in inducing thermal burns by trying to cauterize bleeding points at the skin edge: the best solution is to approximate the skin edges by closing the wound. The key is to get the leg closed before heparin is given, if possible. Good bandaging technique (firm not tight) helps hemostasis (Fig. 7.4).

## Closing the wound

Use 2/0 or 1/0 Vicryl on a curved needle, taking bites 1–1.5cm long and 1cm apart, as shown in Fig. 7.4c, to bring deep fascia together securely. Always sew into the fascial layer, not just the fat which will not support sutures. Avoid button-holing the skin in thin people by making sure that the needle is parallel to the skin. Use a straight handheld, or curved needle with 3/0 Monocryl or Vicryl to close the skin with a subcuticular stitch, making bites of about 1cm, 0.5 cm apart. Avoid putting too much tension on sutures as this causes ischemia: too little may result in the wound separating postoperatively. If this happens use interrupted 4/0 Prolene to repair. If using monofilament put at least six throws on the knots to avoid it unravelling, and bury it to avoid a stitch granuloma.

## Do's and don'ts

***Do's***

- Do handle the vein *minimally*: intimal damage is less.
- Do stay close to the vein: it makes dissecting it out much easier.
- Do use sharp Mayos: crushing the skin impairs wound healing.
- Do preserve the saphenous nerve: it minimizes sympathetic dystrophy postoperatively as well as improving wound healing.
- Do use toothed forceps on skin: non-toothed crush the skin.
- Do ensure good hemostasis: hematomas cause wound infections.
- Do close the skin firmly, but not under tension: healing is better.
- Do evert the skin edges: healing and scars are better.

***Don'ts***

- Don't handle the vein with forceps: they damage endothelium.
- Don't tie to close to the vein: narrowing and kinking results (Fig. 7.4f).
- Don't tie too far away from the vein: clots may form (Fig. 7.4e).
- Don't create skin flaps: cut perpendicularly onto the vein (Fig. 7.4a).
- Don't tether the skin: this makes painful and ugly scars.
- Don't bandage too loosely: hematoma and edema will result.
- Don't bandage too tightly: this causes ischemia and poor healing.

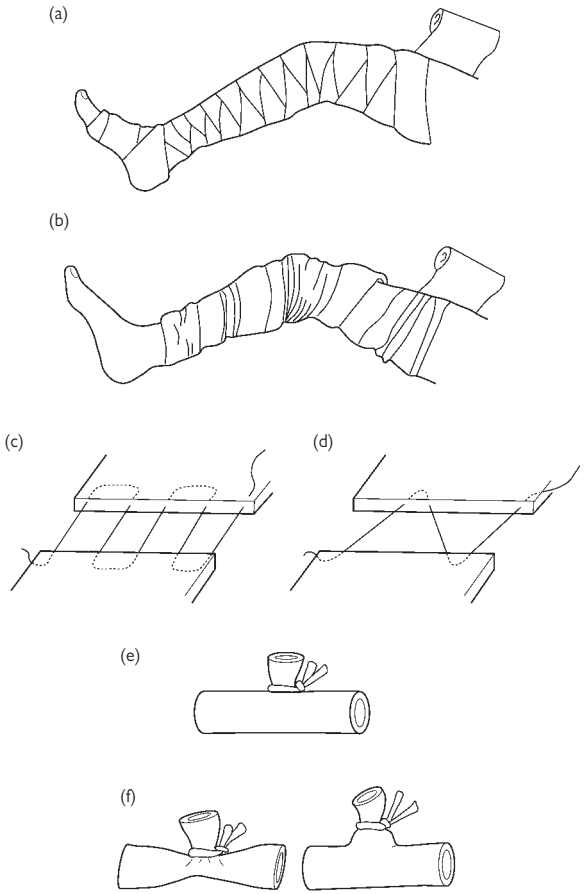



**Fig. 7.4** (a) Correct methods of bandaging. (b) Incorrect method of bandaging. (c) Correct path of skin sutures. (d) Incorrect path of sutures. (e) Correct way of tying side-branches. (f) Incorrect way of tying side-branches.

# Open saphenous vein harvest 5

## Key facts

If the distal LSV in both legs is unsatisfactory in calibre or quality, the saphenous vein in the thigh is dissected out. If this is unsatisfactory, or has already been harvested, the SSV will give one length of conduit from each leg.

## Long saphenous vein harvest in the groin

The landmarks of the LSV in the groin are less constant: the key is to make an oblique incision of adequate length starting over sapheno-femoral junction (SFJ), and extending it slightly medially.

- The SFJ is located at the mid-inguinal point (halfway between bony landmarks of the symphysis pubis and the anterior superior iliac crest).
- Extend the incision from there slightly medially for 8–10cm.
- The LSV is usually located in a plane between layers of fat.
- If it is not immediately obvious, place a self-retaining retractor in the wound, and using blunt dissection with a curved Mayo scissors systematically explore the incision.
- The vein lies within fat: if muscle is exposed you are too deep.
- The vein is most constant at the SFJ.
- Several branches can mimic the LSV: they are either much smaller, or follow courses much more medial or lateral.
- Follow it with the Mayo scissors as described on 📖 p312.
- ***Remember to put the tester in distally, not at the groin end!***
- Leave an adequate stump of saphenous vein to tie proximally so that the femoral vein is not compromised at all by ligatures.

## Short saphenous vein

The SSV runs from 2cm behind the lateral malleolus, over the Achilles tendon, up the back of the calf to the popliteal fossa. The most difficult part of harvesting the SSV is positioning the leg.

- The simplest way to position the leg is to flex the knee, use a towel clip to anchor the foot drapes to the table, and then adduct at the hip exposing the lateral aspect to start, abducting the hip later.
- Identify the SSV distally and then proceed to dissect out the SSV in exactly the same way as the LSV.
- Stop just before the SSV enters the popliteal fossa.

## Using a vein stripper

A stripper is simply a 1cm metal ring, mounted at an angle on a long handle (Fig. 7.5). This method is quicker than conventional harvest as there is no long incision to close, but real care should be taken to avoid avulsing small branches and damaging the endothelium with too much traction. Three lengths of vein can be removed quickly, and with only small skin incisions to close this is a fast technique that improves wound healing and mobilizing. The drawback is a slightly higher rate of avulsed side branches requiring repair on the LSV, and which can cause leg hematomas.

- Expose the saphenous vein at the groin and ligate it proximally.
- Place a 0/0 silk suture distally in the vein and cut it long.

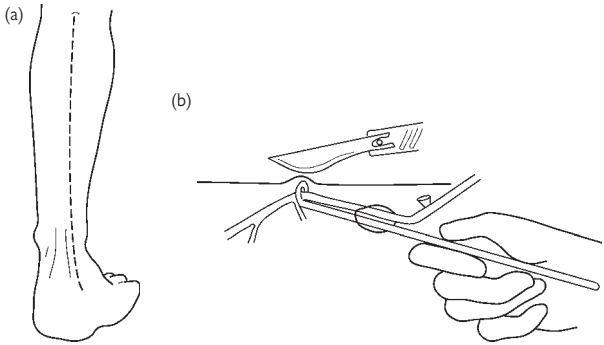



**Fig. 7.5** (a) Short saphenous vein incision. (b) Using the vein stripper to harvest LSV.

- Pass the suture and the vein through the ring end of the stripper and pass the stripper down the vein until you meet resistance with skin tethering caused by a branch.
- Make a 1.5cm incision down onto the stripper where it has come up against a branch, and pass the stripper and vein up through the incision.
- Remove the vein from the stripper and the incision, and ligate and divide the branch, which is usually easily visible.
- Continue as far as necessary and test the vein in the usual manner.

## Endoscopic vein harvest

There are several systems available, e.g., Vasoview, and they offer different approaches to the various aspects of endoscopic vein harvest:

- The saphenous vein is identified just above the knee about two finger breaths posterior to the medial patella.
- The subcutaneous space is created by blunt dissection anterior to the vein initially by digital dissection, then using balloon inflation with gas insufflation to maintain the subcutaneous space.
- Dissection is carried out with long-handled, blunt or 'U'-tipped instruments: side branches are clipped and divided with diathermy.
- Similar techniques have been used for radial artery harvest.
- While a large non-randomized study of endoscopic vs open vein harvest[1] showed lower patency rates in endoscopic vein grafts at 12–18 months, several of the participating centers were low-volume units, and equipment has moved on several iterations. A recent much larger study showed no increase in repeat revascularization at 4 years.[2] Certainly morbidity from leg wounds including infection is much less with endoscopic harvest: and in many countries, particularly with established physician assistant programs (the learning curve is longer than most surgical trainees spend on many units), it is the technique of choice.

## References

1. Lopes RD, Hafley GE, Allen KB, *et al.* Endoscopic versus open vein-graft harvesting in coronary-artery bypass surgery. *N Engl J Med.* 2009;**36**:235–44.
2. Dacey LJ, Braxton JH, Kramer RS *et al.* Long-term outceoms of endoscopic vein harvesting after coronary artery bypass grafting. *Circulation.* 2011;**123**:147–53.

# Radial artery harvest

## Key facts

- The technique of radial artery harvest is completely different than saphenous vein harvest: several important anatomical structures lie very close to the radial artery, which is much less forgiving of poor dissection, handling, and avulsed branches than the saphenous vein.
- Ideally you should wear operating loupes. Forearm anatomy is described on 📖 p54. Assessing conduit pre-op is described on 📖 p172.

## Indications

- Young patients (<65 years).
- Tight stenoses in a left-sided coronary distribution, with good run-off.

## Contraindications

- Poor Allen's test (📖 p172).
- Unsuitable targets (📖 p303).

## Technique

Make sure that the anesthesiologist is aware that radial artery will be harvested, and that they are clear which side: the radial artery catheter must be inserted on the contralateral side.

- Make sure there is an arm-board and an additional diathermy plate.
- Prep and drape the arm, once the patient has been draped (Fig. 7.6).
- Make an incision, just through the dermis, from the radial pulse at the proximal palmar crease to 1cm distal to the brachial pulse, straight or following the hairline (Fig. 7.6) using a 23 blade.
- With the cautery on coagulation at 50 divide the fat.
- Avoid dividing the superficial cephalic vein and the cutaneous nerves.
- Take great care at the wrist where the radial is very superficial.
- Avoid touching the radial artery at all times and keep wetting it with verapamil solution to minimize vasospasm.
- Proximally cauterize down onto the belly of brachioradialis (BR) and divide the filmy fascia that attaches it to flexor carpi ulnaris (FCU).
- If you continue to cauterize along the tendon of FCR, there is minimal risk of inadvertently damaging the radial artery.
- Place a self-retainer carefully between BR and FCU exposing the radial artery in the proximal forearm.
- Using a McIndoe scissors carefully divide the fascia overlying it all the way down to the distal forearm, as far as the proximal palmar crease.
- With the McIndoe scissors perpendicular to the artery, spread the fascia around the artery gently to expose all the side branches.
- Clip them flush with the venae comitantes alongside the radial artery.
- Holding the venae comitantes not the artery itself, divide the side branches on the FCU side using diathermy on 30–40.
- Now lift the radial artery up gently, Ligaclip, and divide the side branches that run beneath it. Then divide those on the FCR side.
- This becomes more difficult in the distal wrist where the plane of dissection is less clear and the radial artery much smaller.
- Continue proximally beyond the first muscular branch to FCR, but stopping at least 1cm distal to the origin of the radial artery.

- Now perform the intraoperative version of Allen's test: place a bulldog clip in the middle of the radial. If you cannot see or feel a pulse distally, or if pulsatile blood does not emerge from a needle placed distally, the collateral palmar circulation is not adequate and you must not divide the artery.
- If the test is satisfactory place an arterial clip proximally and distally, divide the artery with a knife, and place a mosquito clip or marker stitch to mark the proximal end.
- Transfix and tie both radial artery stumps with 3/0 silk or nylon.
- Place a 16G Redivac drain in the wound and secure it.
- Close the deep fascia with 3/0 Vicryl and the skin with 3/0 Monocryl.
- Dress and bandage firmly and place the Redivac drain on suction.
- There should be minimal drainage in the drain at the end of the case.
- The radial artery can be stored in a variety of solutions, e.g., verapamil 30μmol/L heparinized blood: some surgeons prefer the fascia to be divided with a Potts scissors and the artery distended.

**Structures at risk during radial artery harvest**

- Ulnar artery at its origin.
- Median nerve throughout its length.
- Superficial radial nerve throughout its length.
- Lateral antebrachial cutaneous nerve.
- Palmar cutaneous branch of the median nerve.
- Superficial dorsal branch of the radial nerve.

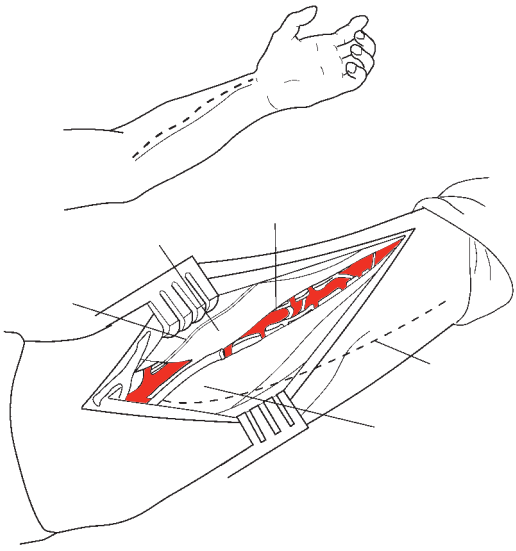


**Fig. 7.6** Incisions for radial artery harvest and structures at risk.

# LIMA harvest

## Key facts

- A LIMA to LAD anastomosis is the most important aspect of CABG.
- Internal mammary artery can be harvested as a ***pedicle*** (with a strip of fascia, muscle, fat, and internal mammary veins intact), or ***skeletonized*** (removed without any other tissues attached).
- The pedicled method, which is used most commonly used, is described here, and the skeletonized technique described on 📖 p320. The LIMA anatomy is described on 📖 p54.

## Indications

Used for LAD, or where RIMA is anastomosed to LAD, the LIMA can go to diagonal or Cx lesions (📖 p321).

## Relative contraindications

- Inadequate flow.
- Salvage surgery.
- Previous radiotherapy to the chest wall.
- Left subclavian stenosis.

## Technique

Good exposure is particularly important. The underside of the chest wall should be just above the surgeon's eye level. Some surgeons sit down to take the LIMA so that the assistants taking other conduit are not disadvantaged by the table being fully raised. After the median sternotomy, with the pericardium open or closed, place the mammary retractor in the incision and open it.

- Open the parietal pleura, or keep it closed by dissecting it off the chest wall (keeping it closed may reduce postoperative pleural effusion, and offer benefits in terms of postoperative lung function).
- If the lung obscures the view you can either pack it down with a large wet swab, or ask the anesthesiologist to reduce the tidal volume.
- Identify the LIMA. This is easiest proximally.
- Note how it curves medially, proximally.
- With the diathermy on 60 dessicate, make one tramline 3–5mm medial to the medial mammary vein, from the manubriosternal junction to the xiphisternum. Some surgeons tramline the lateral side too.
- Turn the diathermy down to 30, and holding on to the parietal tissue with the DeBakeys for countertraction, use a combination of diathermy and blunt dissection with the flat of the diathermy blade to work the LIMA pedicle off the chest wall.
- Side branches are limited to the rib spaces: it is easiest to get into the plane of dissection where the LIMA overlies the ribs.
- Clip the side branches flush with the LIMA and divide them at the chest wall with cautery: some surgeons do not use clips at all.
- Now, using diathermy, divide the parietal pleura on the lateral aspect of the LIMA pleura to free the pedicle from the chest wall.
- Applying countertraction, divide the side branches and fascia so that the proximal portion is freed from the chest wall, to the 1st rib.

- Give full-dose heparin, and check hemostasis.
- 2min after the full dose of heparin is given, place a clamp over the distal stump, divide the mammary at the bifurcation, check that there is good pulsatile flow from the mammary and then place a large Ligaclip over the distal end. Flow often improves during the time between initial harvest and LAD anastomosis.
- Wrap the mammary in a small swab soaked in papaverine, and place it in the pleura for later use. Tie the distal stump with 2/0 Vicryl.

## Tips and pitfalls of LIMA harvest

- If you cannot visualize the mammary, look and feel for pulsations.
- Always maximize exposure: sometimes a few extra turns on the retractor and tilting the table away make a difficult section easy.
- If the LIMA is very stuck, use minimal diathermy: it is easiest to get into the right plane proximally, but safest to do so distally.
- If your pedicle is too narrow you'll spend most time trying to control bleeding from the mammary veins.
- Freeing the mammary proximally, as far as the subclavian vein, gives additional length and reduces the risk of it kinking.
- Packing bleeding and returning 5min later can allow you to see.
- If you can't control bleeding using pressure and fine suction, give the heparin, wait 2min, and place two bulldogs either side of the bleeding point so that you can see more clearly.
- Never clamp the mammary before heparinizing: it may thrombose.

***Structures at risk during LIMA harvest***

Form proximal to distal:

- Subclavian vein (avoid this—it is very difficult to reach to repair!).
- Phrenic nerve proximally where it is close to the LIMA, or more distally when you divide the pleura.

***LIMA too short***

- Skeletonizing, harvesting an additional inch or so proximally, and opening the pleura to drop it under the lung all give more length.
- If you injure the LIMA it is rarely possible to adequately repair it: if the injury is proximal consider using it as a free graft.
- Never leave the LIMA under tension: the graft will fail in the early post-op period, with potentially fatal anterior wall ischemia.

***Poor flow***

- Poor flow. Some surgeons advocate discarding the LIMA if flow <20mL/min, but others will use even low flow as long as it is pulsatile.[1] Low flow may have several causes:
  - Distal to bifurcation, or includes more vasopastic distal portion: just divide proximal to bifurcation and soak in papaverine.
  - Dissection: look for hematoma, and use proximal segment.
  - Spasm: usually improves with topical papaverine.
  - Tip crushed by scissors: use sharp instruments!
  - Avoid injecting papaverine: it has pH <4 and may damage intima.

1. Hata M *et al.* Clinical results of coronary artery bypass grafting with use of the internal thoracic artery under low free flow conditions. *J Thorac Cardiovasc Surg* 2000; **119**:125–9.

# RIMA harvest

## Key facts

- This can be taken as a pedicle, but is more commonly taken skeletonized to avoid compromising wound healing if the LIMA has also been harvested.
- An additional reason for taking it skeletonized is the extra length needed to reach most potential targets.
- Anatomy of the RIMA is described on 📖 p58. Patient selection is described on 📖 p302.

## Indications

Any patient undergoing CABG.

## Contraindications

- Avoid bilateral mammary harvest in obese, diabetic females as it increases the risk of deep sternal wound infection (DSWI).
- Severe COPD, as BIMA and cough predisposes to sternal non-union.

## Technique

Reverse the mammary retractor, so that it can be used on the right. If the LIMA has been taken it, do not divide it.

- Open the pleura and identify the RIMA.
- With the diathermy on 25–30, make a small nick in the parietal pleura adjacent to the mammary vein.
- Use a combination of blunt dissection with the flat of the blade and diathermy, free an area of pleura from the mammary, and continue the length of the RIMA. The easiest way to do this is to pull down very firmly on the pleura where it has been opened up and then use the diathermy in as short bursts as possible to extend the incision.
- The underside of the length of the RIMA should be left exposed.
- If you do this correctly you should hardly see the mammary veins.
- Now, still using blunt dissection, extend the plane between the RIMA and the chest wall: the best way of doing this is to pull very gently on the adventitia that is left on the underside of the RIMA and use the flat of the diathermy blade (without using the cautery) to work between the upper side of the RIMA and the chest wall.
- The best place to start this dissection is overlying the ribs where there are no side branches (side branches occur in the intercostals spaces).
- Place small Ligaclips flush with the RIMA and the chest wall on individual sidebranches, and divide the branches with a Potts scissors.
- Again, more care is required as it is easy to avulse the side branches.
- Continue so that the RIMA is free from xiphisternum to the 1st rib.
- Give heparin.
- 2min after the full dose is given, divide the mammary, check the flow, ligate both ends, and wrap the RIMA in a small swab soaked in papaverine.
- Incise the pleura so that the RIMA can be brought out to the RCA or through the transverse sinus to the left-sided coronary circulation.

### Pitfalls in right internal mammary harvest

- Avoid damaging the mammary veins. Clipping them causes them to become congested, and obstruct the view even more.
- The RIMA has a shorter proximal course than the LIMA (📖 p58).
- Always take the RIMA to the bifurcation: the maximum length of RIMA is almost always used.

## Total arterial revascularization

Total arterial revascularization (TAR) eliminates vein grafts, using various combinations of LIMA, RIMA, radial artery, and occasionally gastroepiploic artery to perform CABG. The rationale is driven by the observation of superior patency of arterial grafts compared to SVG (📖 p302). A large number of configurations are used, primarily reflecting constraints of length and availability of arterial conduit; the impact of target (distal run-off and lesion severity) on radial artery patency; competitive flow; and the desire to eliminate the need for aortic manipulation (anaortic or 'no touch' technique), to minimize the risk of stroke.

***Anaortic or 'no-touch' technique***

- This is associated with mortality and stroke rates <0.5% in several series, and may be the strongest reason for OPCAB versus ONCAB.
- Surgery is off-pump to eliminate need for aortic cannulation.
- Proximal aortic anastomoses are eliminated by:
  - Using the *in situ* RIMA to graft the RCA and proximal PDA (this can be a struggle for the *in situ* RIMA to reach without tension), or the LAD (with LIMA being used for Cx lesions), or the proximal Cx and OMs via the transverse sinus.
  - Using RIMA as a 'T' graft off the LIMA allowing multiple sequential grafts of more distal circumflex and PDA territories. While this allows maximum versatility, the risk of compromising the LIMA to LAD anastomosis is not negligible.
  - Using radial loop grafts off the *in situ* RIMA to reach distal targets.

***Evidence supporting this approach***

Although patency rates superior, evidence of resultant clinical benefit less impressive. It's unfortunate that none of the largest registry studies or RCTs have focused on anaortic TAR, as this likely provides the most benefit in terms of freedom from death or stroke after revascularization. The most widely cited study in support is the Cleveland Clinic 10000 patient registry study[1] which does not control for multiple key confounding variables, or identify which reoperations were for graft failure as opposed to valve surgery, progression of native disease, etc.), and does not show survival benefit in several patient subgroups. There are several RCTs: 10-year results expected in 2015.

***Reasons for low uptake***

TAR represents <15% of CABG in US, because it is a technically more demanding and time-consuming procedure, with short-term complications more compelling in clinicians' minds than unclear long-term benefits.

## Reference

1. Lytle BW *et al.* Two internal thoracic artery grafts are better than one. *J Thorac Cardiovasc Surg* 1999; **117**:855–72.

# Alternative conduit

## Key facts

- The gastroepiploic artery as a pedicled graft and the inferior epigastric artery as a free graft have both been used as arterial conduit when more conventional options are not available.
- The contraindications to using these conduits are abdominal surgery, previous radiotherapy in the area, severe peripheral vascular disease, or evidence of end-organ ischemia.
- Cephalic vein and splenic artery have been used rarely.

## Right gastroepiploic artery harvest

- Pass an NG tube to decompress the stomach.
- Extend the median sternotomy incision, dividing the linea alba from the xiphoid to halfway above the umbilicus.
- Divide the triangular ligament of the liver, and retract the liver superiorly and to the right (Fig. 7.7) and pull the stomach into view.
- The right gastroepiploic artery (RGEA) runs along the greater curvature of the stomach from the duodenum towards the cardia.
- Dissect off the RGEA clipping and dividing the side branches in the way described for the radial artery, working from the pylorus until sufficient length is achieved.
- Inadvertent diathermy burns to the stomach mucosa should be oversewn with non-absorbable monofilament, e.g., Prolene.
- Make an opening in the diaphragm medial to the IVC.
- Pass the pedicle either anterior or posterior to the duodenum and stomach, depending on where there is least distortion, through the diaphragmatic opening into the pericardial cavity.
- The RGEA is normally anastomosed to the distal RCA or PDA.

## Inferior epigastric artery harvest

- The right inferior epigastric artery should be harvested if the LIMA is used, to avoid ischemic changes to the skin.
- Make a paramedian incision beginning at the level of the umbilicus along the lateral border of rectus abdominus, ending 2–3cm above the inguinal ligament (Fig. 7.7).
- Divide the anterior rectus sheath and retract the muscle laterally.
- Mobilize the artery with the venae commitantes as described for the radial artery (Fig. 7.6).
- Divide the artery and its pedicle proximally at its origin from the external iliac artery where it may be calcified, and distally where it bifurcates.
- Mark the distal end with a marker stitch or hemostat.
- Store it in a solution of verapamil in 30μmol/L heparinized blood until it is required.

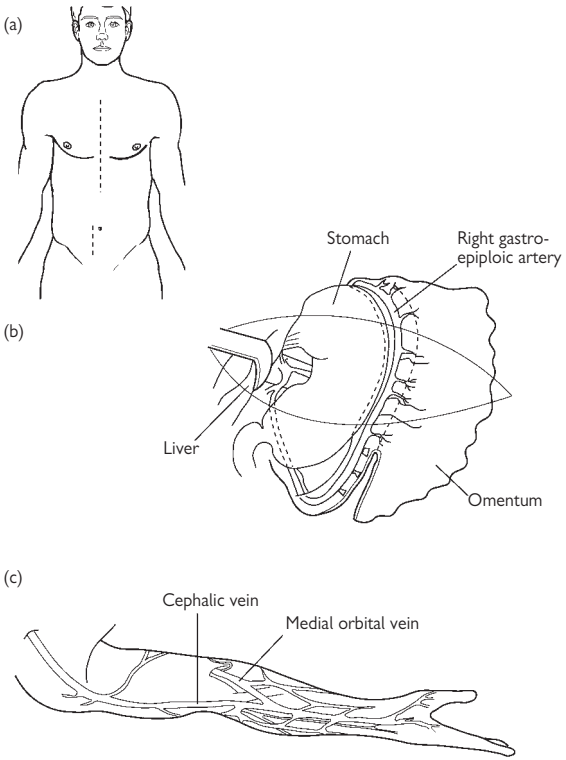



**Fig. 7.7** (a) Incisions for right gastroepiploic artery harvest (RGEA) and right inferior epigastric artery harvest. (b) Exposure of the RGEA. (c) Cephalic vein.

## Splenic artery

The splenic artery is exposed through a midline extension of the median sternotomy, as described for the RGEA. The lesser peritoneal sac is opened to gain access to the artery which travels along the superior margin of the pancreas. The spleen has a collateral blood supply from the left gastric artery, so splenectomy is not required.

## Cephalic vein

- The cephalic vein can be harvested through an incision running from the anatomical snuff-box, over the lateral aspect of the forearm to the bicipital aponeurosis, using the open technique described for saphenous vein harvest (📖 p306).

# Conduct of bypass for CABG

## Key facts

There are a few aspects of cardiopulmonary bypass specific to CABG.

## Heparinization

It is important not to divide and ligate the LIMA (and RIMA if that is also being harvested) before the patient is fully heparinized, as there is a risk that thrombus could occlude the lumen. Most surgeons give heparin just as they are finishing the most proximal part of the mammary dissection and spend the 2min after the heparin has being given checking mammary bed hemostasis

## Aortic cannulation

The key is to allow enough room for the cross-clamp, cardioplegia cannula, and all top-ends. If the ascending aorta is short, or multiple top-ends are planned, free the ascending aorta as far as the proximal arch. Placing a sling around the aorta allows gentle traction to be placed on the aorta bringing more length into view. Take a McIndoe scissors and divide the adventitia lying between the aorta and the PA trunk. Place your index finger and thumb around the aorta, and extend the plane of dissection with blunt dissection. When minimal or no tissue is palpable between your fingers pass a Sem C-clamp from right to left underneath the aorta and pull a wet tape through. Place this on gentle traction by clipping the tape underneath the horizontal bar of the sternal retractor.

## Venous cannulation

A two-stage cannula or a Ross basket is sufficient, unless combined mitral valve surgery or VSD surgery is planned, in which case venous cannulation should be bicaval.

## Cardioplegia

Cardioplegia is given anterogradely. Retrograde delivery is useful when:

- Tight proximal stenoses mean that antegrade infusion may not deliver complete cardioprotection, or to maximize cardioprotection in the very hypertrophied, ischemic, or dysfunctional LV.
- Cardioplegia is given antegradely initially, and then retrogradely at intervals. Once each distal anastomosis is performed cardioplegia may be given directly down the vein graft as well as retrogradely.
- If you are using a single-clamp technique, you can complete proximal anastomoses while giving retrograde.

## Venting

The heart is normally vented by either an aortic root vent, usually a 'Y' connector on the cardioplegia cannula, or by an LV apical vent. When vessels are occluded aortic root venting may not be enough to keep the coronaries free of blood during the distal anastomosis.

## Single clamp technique

Clamp time may be minimized by completing all distals first, removing the cross-clamp, and applying a side-biting clamp for the proximal

anastomoses. In the single-clamp technique the cross-clamp remains *in situ* and a side-biting clamp is not used for the proximals which can be performed after each distal anastomosis, or at the end prior to removal of the clamp. The primary advantage of the single-clamp technique is less risk of stroke from repeated manipulation of the aorta.

## Alternatives to cardioplegic arrest on bypass

As the heart remains perfused these techniques mean that it is sensible to try to restore flow to the areas affected by the most important stenoses first. The top-ends are usually performed after (or in some cases before) each distal anastomosis, and the heart allowed a period of perfusion before moving onto the next distal anastomosis.

### *Beating heart on bypass*

Cardioplegia is not given, and the heart is not arrested. Instead stabilizers used in OPCAB are used to position the heart once it is on bypass, and distal anastomoses carried out in the normal fashion (📖 p328). Useful where aortic calcification precludes clamping but not cannulation.

### *Intermittent cross-clamp fibrillation*

This is one of the oldest forms of cardioprotection, popular before the development of potassium-based cardioplegia. An increased understanding of ischemic preconditioning (📖 p75) and the adverse effects of cardioplegic arrest (📖 p74) have led to revived interest in this technique. The principle is based on the fact that a brief period of ischemia (<20min of cross-clamping) is readily reversed by perfusion with normal blood. The few randomized studies comparing this technique with cardioplegic arrest show no difference in mortality, stroke, or postoperative indices of ventricular function. Repeated cross-clamping of the aorta does not appear to be associated with increased risk of stroke in elective patients with no risk factors for peripheral vascular disease.

*Indications*: there is little objective evidence to support specific recommendations: it is a safe and useful alternative to cardioplegic arrest. The surgeon must, however, be able to reliably complete a distal anastomosis in <15min.

*Technique*: cardioplegia is not given, and the heart is fibrillated so that the distal anastomoses can be performed. This is carried out under moderate hypothermia (32°C). To prevent LV distension the aorta must be cross-clamped as soon as the heart is fibrillated. The sequence is:

- Fibrillator applied to heart.
- Flows down.
- Cross-clamp on and flows back up.
- Heart positioned.
- Distal anastomosis performed: must take <15min.
- Fibrillator removed as anastomosis completed.
- Anastomosis tested.
- Heart defibrillated.
- Flows down.
- Cross-clamp off and flows back up.
- Proximal anastomosis performed with side-biting clamp.

# Positioning the heart on bypass

## Key facts

- Positioning the heart is key to performing efficient and effective distals.
- It's always easier if the heart is adequately drained—always check the RA and RV are decompressed and your vent is on before lifting the heart, and again if the distal coronaries are obscured by blood flow.

## Positioning the heart for the LAD and diagonals

- Slide your right hand under the heart, palm up.
- Lift the ventricles and rotate them slightly anticlockwise so that the LAD comes into view. This maneuver can be used before bypass to inspect the vessels without too much hemodynamic compromise.
- With your left hand use a DeBakey forceps to pack one or two large wet packs underneath the heart, holding it in position.
- Stay sutures should not be necessary.

## Positioning for intermediate and OM vessels

- Slide your right hand under the heart palm up.
- Swap to small dry pack in your left hand, hold the apex firmly, and lift the heart right up, twisting the ventricles anticlockwise.
- Now pack one or two large wet packs underneath the heart using a DeBakey's in your right hand, and ask your assistant to place their right hand, thumb down across the left ventricle holding it in place.
- If you have two assistants, one can hold the heart from the left side of the table, standing to your left side at the patient's right shoulder.
- Letting the right pleura down sometimes helps exposure.
- Positioning the table head up, and rotating it towards you also helps.
- Inspecting the OM system before bypass is possible, but it causes rapid hemodynamic compromise, and most surgeons do not do it routinely.

## Positioning the heart for the RCA vessels

- Using a DeBakey tuck a small dry pack down behind the LA appendage into the oblique sinus.
- With your left hand pull the acute margin of the heart slightly towards the head so that you can see the tip of the pack.
- Using the DeBakey drag the pack towards the IVC; this rotates the heart exposing the distal right and PDA vessels.
- An assistant can hold the acute margin using a small dry pack.
- Sometimes stay sutures are necessary to improve access.
- Positioning the table head down and rotating it away from you helps.

## Stay sutures

Superficial stay sutures can be used simply to hold the vessel walls open and retract epicardial fat. Deeper stays straighten out the vessel and occlude flow. Superficial stay sutures, normally 3/0 Ethibond, are placed in the epicardial fat parallel to the arteriotomy, and fixed with rubber-shod clips. Deeper 1/0 or 2/0 Prolene is passed perpendicularly into the myocardium 1cm away from the coronary artery and out on the other side, and anchored with clips.

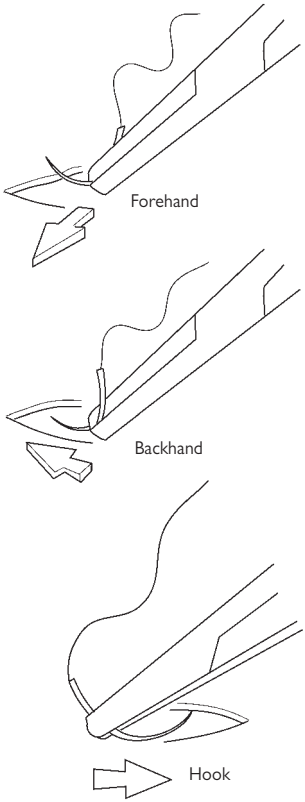



**Fig. 7.8** Different ways of mounting the needle in the needle driver needed for coronary distals.

# Vein distal anastomosis

## Key facts

- Flaps or asymmetry in either the vein or the artery incisions can reduce the lumen of the anastomosis, predisposing the graft to failure. Handling the intima of the vein or artery predisposes to thrombosis.
- Although there is large variation between surgeons, generally the distal anastomoses are performed before the proximal anastomoses, in the following order: distal right coronary artery (RCA)/posterior descending artery (PDA)/lateral ventricular branches (LVB), followed by the obtuse marginal (OM) vessels from distal to proximal, followed by the intermediate vessel (Int), then the diagonal vessels (D) and finally the left anterior descending (LAD).
- There is also a wide variation in techniques of anastomosis. One simple technique is described here for end-to-side anastomosis: a variation on the same technique is described on 📖 p330.

## Technique

Double-ended 8/0 or 7/0 Prolene is used for the anastomoses unless the coronary vessel is very calcified, in which case 6/0 Prolene gives a needle with slightly greater durability. Fine microforceps and a ratcheted Castro needle holder are used (Fig. 7.8).

- Position the heart (📖 p326). Using a fine forceps to retract epicardial fat with your assistant retracting in the same way on the other side of the vessel, clear the vessel of fat and adventitia using a 15 blade.
- Carefully identify the centre of the vessel and, while lifting up on the epicardial fat to separate the back wall from the front wall, open the vessel carefully using either a 15 or an 11 blade, Fig 7.9 (a), avoiding trauma to the back wall (some surgeons turn off vent to minimize this).
- Extend the incision in both directions using a forward Potts followed by a backwards Potts scissors, so that the apexes of the incision are cleanly cut and the incision is 2/3 larger than the vessel diameter.
- Prepare the vein by cutting off the proximal tip obliquely and then incising the heel by about 2mm, so that it matches the arteriotomy, Fig 7.9 (b).
- Suspend the vein by the toe using either a mosquito clip or a fine DeBakey in your assistant's left hand. They follow the suture with their right.
- Place a rubber-shod bulldog clip on the free needle.
- Start two stitches from the heel, pass the needle outside to in on the vein, backhand. Your next stitch is inside to out on the arteriotomy two stitches from the heel, backhand (Fig 7.10).
- Continue outside to in on the vein moving clockwise, inside to out on the artery moving anticlockwise until you are one or two stitches past the heel, on the outside of the arteriotomy.
- Wet the sutures with saline and gently pull on the both ends of the suture while your assistant brings the vein down with slight countertraction onto the arteriotomy, until the suture is taut and

all loops are taken up. If the arteriotomy is very friable put in fewer stitches before bringing the vein down.

- Now holding the vein taut so that the toe of the vein meets the toe of the arteriotomy, pass the needle outside to in on the vein and inside to out on the artery, forehand. The first stitch should be done in two passes, but after that it is possible to stitch both vessels in one pass.
- At the toe change the needle to backhand, and suture the toe in two passes, to ensure that you do not catch the back wall of the artery. A hook shot is an alternative (Fig. 7.8).
- Continue forehand to the start of the anastomoses: the first stitch after the toe is usually easier in two bites, but after that most stitches can be done in one pass—just retract the epicardium so you can see your stitches are full thickness and not catching the back wall.
- Take up the slack in the anastomosis without purse-stringing it.
- Tie the anastomoses: use a slip-knot and then lock it with eight throws.
- If you break the suture while tying it, and there is not enough left to do an instrument tie, the safest thing is to redo the anastomoses.
- Test the anastomosis by injecting heparinized blood, saline, or cardioplegia down the graft, noting how much resistance (there should not be much if the target vessel is >1.5mm), checking that you can see the artery and the myocardium perfusing distally and noting any leaks.
- Take particular care when repairing leaks at the toe, as catching the back wall of the artery here will effectively occlude the anastomosis.
- It is easiest to anastomose the toe before the heel when carrying out PDA grafts: start inside to out on the vein forehand, outside to in on the artery, moving down towards the heel on the diaphragmatic side of the anastomosis, then change needles and complete the toe backhand working anticlockwise forehand to complete the anastomosis.

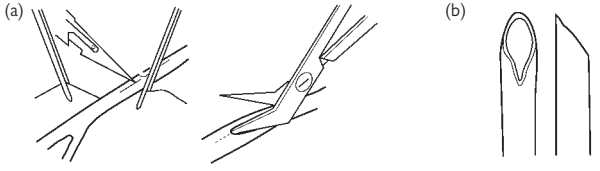



**Fig. 7.9** (a) Creating the arteriotomy. (b) Preparing the vein.

# Arterial distal anastomosis

## Key facts

- The vein can be stretched or gathered slightly to match the size of the arteriotomy, but arterial conduit is not so forgiving, and the size match needs to be exact.
- To achieve this most surgeons do not prepare arterial conduit in the same way as venous conduit: they leave a large hood of excess artery which can be cut to size halfway through the anastomosis. The technique described here is for arterial grafts, but it can just as easily be used for venous grafting (but there is no need to shape the hood).

## Technique

***Preparing pedicled LIMA:*** make a V-shaped incision in the pleura, taking care to avoid the phrenic nerve. Bring the LIMA through from the pleural space and make sure that it is not twisted. Measure the length against where you want to graft the LAD and divide the LIMA accordingly. Blood flow should reflect the perfusion pressure: there should be a strong, horizontal stream of blood. If this is not the case it is advisable to use other conduit. Anchor the LIMA, so that the shiny parietal pleura is uppermost, by clipping it to the side towels. Using 90° scissors divide the fascia in front of the LIMA for 1–1.5cm. Next, staying in the midline, open the front of the LIMA. If the lumen looks particularly narrow, or if there are multiple side branches, continue a little lower.

***Preparing skeletonized RIMA:*** cut the RIMA flush with the Ligaclip to get maximum and check the flow as earlier. Divide the front of the vessel in the midline as described for the LIMA.

***Preparing the radial:*** either cut the distal end obliquely and incise it just as described for the saphenous vein, or create a hood that can be shaped to size later as described for the LIMA

- Distal anastomosis technique suspend the arterial conduit a few inches from the coronary artery.
- Make the arteriotomy as described on 📖 p328.
- Using double-ended 8/0 Prolene pass the first stitch forehand inside to out on the heel of the conduit, forehand outside to in on the arteriotomy, bring the arterial conduit down after.
- Place one needle on a rubber-shod bulldog, and then do three or four stitches forehand on the farside of the arteriotomy inside to out on the conduit, outside to in on the coronary artery, finishing outside on the conduit.
- Next swap the rubber-shod bulldog onto the other needle, and suture the other side of the arteriotomy backhand.
- Continue, swap to forehand, round the toe, just as described on 📖 p329, to the beginning of the anastomoses and tie it down.
- Check the anastomosis by either injecting the conduit with blood, or in the case of pedicled conduit, releasing the bulldog clamp temporarily. Replace the clamp or all cardioplegia will rapidly be washed out.

## Troubleshooting the bleeding distal

- It's a bit depressing to find your distal anastomosis leaks, but it should be possible to sort it out without compromising the anastomosis.
- Always recheck distals with the clamp off, before weaning from bypass, as they can be difficult to fix with the heart full and beating.
- If bleeding is too much to clearly see, you can occlude the graft with a bulldog, or go down on the pump flow for 1min at a time.
- Position the heart as for the distal anastomosis, and squirt saline over the anastomosis to localize the bleeding point.
  - Side branch: usually safest to fix with a figure-of-8 stitch using 8/0 Prolene as a Ligaclip may distort the anastomosis, and if you need to remove it for any reason can destroy the integrity of the hood.
  - Dog-ear: fix poor spacing with 7/0 Prolene figure-of-8 full-thickness bites on conduit and arteriotomy.
  - Loose suture: use a fine nerve hook to carefully pull up the suture (which should stop the bleeding) and pass a 7/0 Prolene through the loop. Take a partial thickness bite adjacent to it across the conduit and arteriotomy taking care not to narrow the anastomosis with a fresh 7/0 Prolene and tie securely. Now tie the free end of this 7/0 to the 7/0 in the loose loop, so that the loop forms part of the knot (Fig. 7.10) which will secure the whole anastomosis.
  - Tear: do a partial thickness figure-of-8 bite using 8/0 Prolene parallel to the anastomosis in the conduit then in the coronary and tie.
  - Sometimes taking the tension off the anastomosis by suturing the divided the epicardial fat together, without constricting the anastomosis is sufficient.
- Deciding whether to give protamine is a fine judgment: often times bleeding is due to needle holes and raw surfaces and resolves, but if it doesn't you may need to use an off-pump stabilizer to get access.

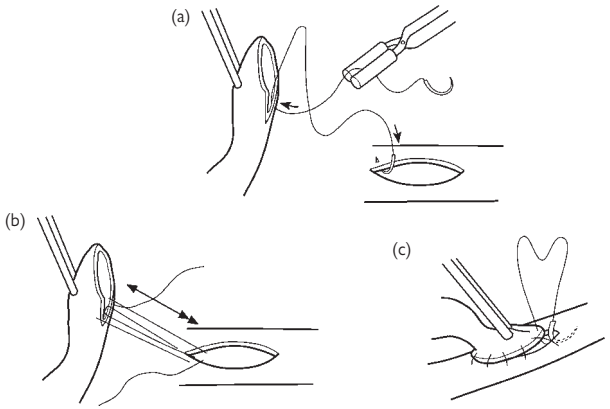



**Fig. 7.10** (a) First stitch. (b) Parachuting the vein down. (c) Stitching the near side.

# Jump or sequential grafts

## Key facts

- Sequential grafts are useful when there is insufficient conduit to graft every artery individually. Any conduit, including LIMA can be used.
- The most distal anastomosis should be to the target vessel with the best outflow; otherwise preferential flow to the proximal anastomosis puts the distal anastomosis at high risk of failure.
- Avoid mixing arterial and venous conduit in 'T' or 'Y' grafts: competitive flow down the vein can cause early failure of the arterial conduit.

*Advantages*

- Using sequential grafts increases the number of arteries that can be revascularized by a single length of conduit from one to two or three.
- The number of proximal anastomoses and hence the need to handle the aorta is reduced, and may even be eliminated by using the IMAs to sequentially graft all targets.
- Patency rates of venous grafts may be increased, because of the augmentation of outflow.

*Disadvantages*

- Larger regions of myocardium are jeopardized if the conduit fails.
- LIMA to LAD flow may be compromised by 'Y' or 'T' grafts proximal to the LIMA to LAD anastomosis.

## Technique

Take your time to measure and orientate the distended conduit correctly with the heart full: it may help to use a marker pen to mark the conduit where you plan to open it. Open the conduit transversely if it lies across the coronary artery, or longitudinally if it will lie parallel to the coronary artery (Fig. 7.11). For transverse incisions, the length of the incision should not exceed 2/3 of the diameter of the conduit, or the conduit will be flattened by the anastomosis. If using IMA or radial skeletonize the portion to be used in the side-to-side graft.

- Position the conduit 1cm away from the arteriotomy in the position that it will eventually lie in.
- Starting one stitch away from where you want the conduit to meet the heel of the arteriotomy, pass the suture from outside to in on the conduit and place a rubber-shod bulldog on the other end.
- Now suture inside to out on the arteriotomy, two stitches from the heel.
- Continue two stitches past the heel, then parachute the conduit down gently, check that there are no crossed stitches or loops, and take up all the slack in the suture.
- Most of the following stitches can be done in one pass: continue with the same suture along the nearside of the vessels to the toe and finishing on the far side.
- It is sometimes easier to start side-to-side anastomosis in the way described on 📖 p328: i.e., suturing outside to in on the conduit, inside to out on the arteriotomy at the heel, and tying down immediately.
- If the conduit is small, a transverse incision reduces the risk of the anastomosis constricting it.

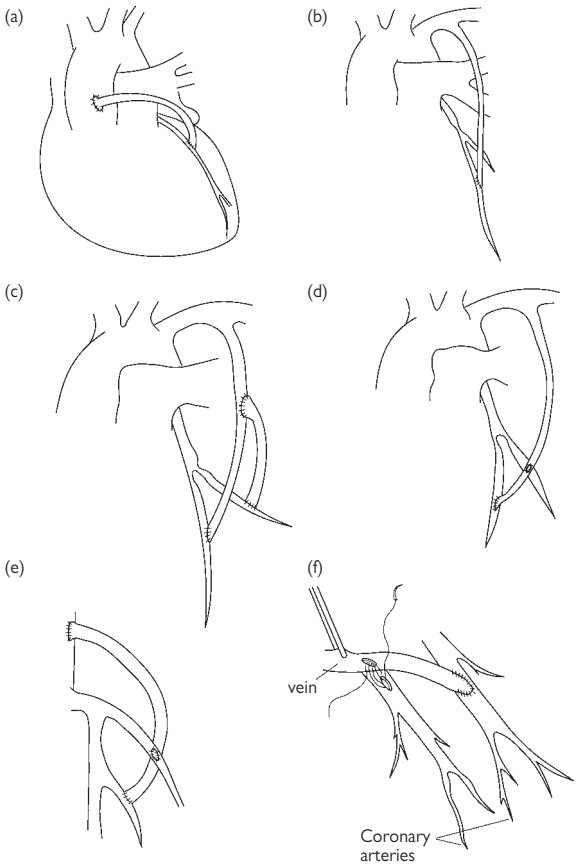



**Fig. 7.11** Types of bypass graft. (a) Vein graft to the LAD, proximal anastomosis to aorta. (b) LIMA to LAD. (c) 'T' graft: LIMA to LAD with free RIMA anastomosed to LIMA proximally and diagonal distally (common combination is the RIMA anastomosed to diagonal or OM with PDA as a sequential. (d) Sequential graft LIMA to LAD and diagonal. (e) Sequential vein graft to LAD and diagonal with proximal anastomosis to aorta. (f) Technique of performing sequential vein graft.

# Endarterectomy

## Key facts

- Endarterectomy was one of the early surgical treatments for coronary disease, but as results were poor it was largely superseded by CABG.
- If a vessel is completely occluded but has a large distribution supplying ischemic and viable myocardium, an endarterectomy may be indicated (probably not more than 2–5% of CABG cases).
- Arteries that can be seen on angiography generally do not require endarterectomy as they have a true lumen that can be grafted: endarterectomizing them may result in a perioperative MI if acute thrombosis occludes the endarterectomized vessel.
- Any conduit may be used, but as patency of endarterectomized vessels is poorer than non-endarterectomized vessels, the IMA may be better employed elsewhere.
- Endarterectomy patency rates are best in the largest vessels, with large territories: the RCA is the most commonly endarterectomized vessel.

***Advantages***

Grafting is possible in vessels that it would otherwise be impossible to graft.

***Disadvantages***

- Endarterectomy has been shown to be an independent risk factor for increased morbidity and mortality postoperatively. This may reflect patient selection, technique, and perioperative management.
- In the longer-term, grafts to endarterectomized vessels have lower patency rates than grafts to diffusely diseased but non-endarterectomized vessels.
- Difficult to do safely off-pump as the arteriotomies are often very long.

## Technique

- The anterior wall of the artery is opened in the normal way with a 15 blade.
- The creamy-yellow, firm atheromatous core is gently separated circumferentially from the vessel wall using a curved dissector (Fig. 7.12).
- Maintaining gentle tension on the core, without breaking it off, the plane between the vessel wall and the core is developed and the core teased out proximally and distally.
- Usually it is possible to extract the atheroma for several centimeters as far as the distal branches, and it should be seen to taper—confirming that distal branches have been adequately endarterectomized.
- Once the atheroma has been removed, conduit can be anastomosed to the vessel walls in the usual manner.
- If the endarterectomy breaks off, it may be necessary to extend the arteriotomy far enough to continue the endarterectomy: a vein patch may be needed to close a long arteriotomy which can then be grafted with arterial or venous conduit.
- Some surgeons routinely perform a long endarterectomy with vein patch.

## Postoperative management

Because of the damage to the coronary intima, these anastomoses are much more prone to acute thrombosis postoperatively (5% risk of postoperative MI in recent series, down from up to 25% in historic series) and aspirin should be given early. These patients are either anticoagulated for 2–3 months postoperatively, or maintained on aspirin and clopidogrel. Some surgeons start heparin as soon as mediastinal drainage is <50mL/h.

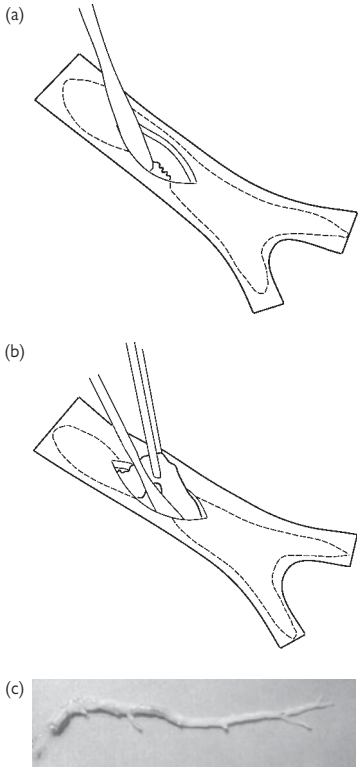



**Fig. 7.12** (a) The endarterectomy plane is developed. (b) Atheroma is teased out. (c) Endarterectomy specimen—note tapered ends. Fig 7.12c reprinted with permission from Fukui, T, Takanashi, S, Hosoda, Y (2005) Coronary endarterectomy and stent removal in patients with in-Stent restenosis, *Ann Thorac Surg*;**79**:558–63. Copyright © The Society of Thoracic Surgeons.

# Positioning the heart in OPCAB

## Key facts

- The LAD and diagonal vessels are easiest to access, the RCA can cause problems with bradycardia, and the Cx vessels are more challenging to access particularly if close to the AV groove.
- Positioning the heart for off-pump surgery is a balance between achieving good exposure and maintaining hemodynamic stability.
- Certain maneuvers, particularly compressing the long axis of the heart, cause AV valve incompetence. The heart tolerates lifting and rotating well as long as compression and distortion are avoided.
- Opening the pleura widely improves positioning as the heart can be rotated further without compressing it, and **gradual movements** allow the heart to compensate.
- Reflex tachyarrhythmias may develop as a result of low CO: this makes the distal anastomoses much more difficult, and along with extrasystoles and rising PA pressures are an early warning of ischemia.
- Communication between surgeon and anesthesiologist is key to successful off-pump surgery (📖 pp98–99).

## Exposure

***Incision:*** usually a sternotomy for multi-vessel grafting, although left and right thoracotomy approaches are also used for more limited grafting. A limited skin incision is cosmetically appealing but reduces exposure and increases the amount of manipulation of the heart required for good access particularly for PDA grafting.

***Pericardium:*** this is widely incised, with an extended 'T' inferiorly, extended down to the IVC, avoiding the phrenic, to maximize exposure. Placing a line of three stay sutures parallel to and just above the left phrenic. and pulling up on them (place the in the self-locking OPCAB retractor) brings the whole anterior surface of the heart completely into view.

***Tape and snare:*** a broad tape (or a small pack) (Fig. 7.13b), is fixed to the posterior pericardium between the right and left superior pulmonary veins with a 0/0 Ethibond, which is then snared. Traction on the tape and the snare serves to lift the apex of the heart, and is particularly useful for Cx and PDA territory grafts.

## Stabilizers

There are a variety of stabilizing adjuncts.

***Suction devices:*** there are two types. The first is a cup-shaped device that is placed on the apex of the heart to position and stabilize. The second is a two-pronged foot, the prongs being placed either side of the segment of artery being grafted, to stabilize that region of heart. Both suction devices have a flexible arm that can be adjusted, and then tightened to make it rigid, and clips by which then can be attached to the sternal retractor. Some systems require a custom retractor. The connection tubing to the suction has a tap that can be turned on and off by the surgeon. The devices may be disposable, but reusable ones are available.

***Pressure stabilizers:*** these are simple rigid systems that can be set up so that a metal plate stabilizes the area being grafted by pressing down on it. They do not stabilize as well as the suction devices, and tend to cause more hemodynamic compromise. Because they are less bulky than suction stabilizers, these devices have more of a role in mini-thoracotomy coronary surgery, and they are reusable so reduce cost.

### Pitfalls of using stabilizers

- It is possible to damage the descending aorta and esophagus if stay sutures are placed too deep in the pericardium.
- The phrenic nerve is at risk when opening the pleura widely.
- Compressing and distorting the heart reduces cardiac output: although blood pressure may be maintained with pressors, the patient becomes increasingly acidotic.
- Too high suction causes myocardial bruising.

### Positioning the heart for LAD and diagonal vessels

Lift the ends of the tape either side of the heart up and left towards the head, pull the snare downwards, and apply the pronged suction to the segment of artery that is to be grafted (base on lower bar of retractor).

### Positioning for the intermediate and OM vessels

Lift the ends of the tape upwards and to the left, allowing the heart to rotate to the right into the right pleural space, facilitated by opening the right pleura widely. It is usually necessary to stabilize the apex, and rotating the table to the right usually improves access. Apply the pronged suction (base usually on lower bar of retractor).

### Positioning the heart for distal RCA and PDA

Lift the tape up and to the left, stabilize the apex with a suction cup and rotate the table towards the left and put it head-down.

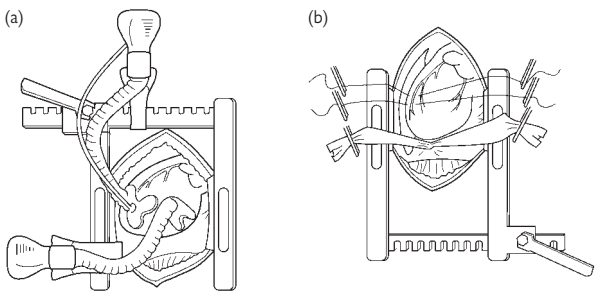



**Fig. 7.13** (a) Positioning the heart for the LAD using two suction devices. (b) Positioning the heart for the LAD using stay sutures and a pericardial sling.

# Distal anastomoses off bypass

## Key facts

- The distal anastomoses may be performed before the proximal ones or after: measuring the conduit when the proximal end has been performed first is slightly more difficult, but the advantage is that blood supply is immediately restored once the distal anastomosis is complete.
- It is usual practice to graft the LAD first to restore blood supply to the anterior wall and septum. Occasionally if the LAD provides the sole collateral blood supply to an occluded RCA it may be preferable to graft the RCA first, to reduce ischemia when the LAD is being grafted.
- The same suture techniques described on 📖 p328–31 can be used in off-pump surgery, although taking bites inside to out on the coronary is easier than outside to in if a shunt is used.
- Early recognition of ischemia (rising PAPs, extrasystoles, bradycardia, and ST-segment changes) is crucial to allow adjustments to be made, avoiding critical decompensation.
- Do not inject saline down completed vein grafts: its acidic pH can cause arrhythmias and hemodynamic disturbance.

## Shunts

Using a shunt diverts blood from the operative field while ensuring that coronary blood flow is disrupted for the minimum amount of time. This is particularly valuable when performing the anastomosis to the RCA as ischemia in this territory leads to bradycardia and hypotension. If the lumen of the coronary artery is >1.2mm in diameter, a disposable shunt can usually be inserted once the arteriotomy has been created. The shunt consists of a flexible 1–2cm round-ended tube, with a tag that ensures that it doesn't migrate into the coronary vessel (Fig. 7.14). They are usually available in a range of sizes (1.0mm, 1.5mm, 2.0mm, and 2.5mm).

- Using two fine forceps, insert one end into the proximal part of the arteriotomy, then bend the shunt so that the other end can be inserted into the distal part of the arteriotomy.
- Clamping the shunt with the forceps as you do this occludes the blood flow, improving visibility.
- Perform the anastomosis normally, taking care not to inadvertently suture the shunt—the thin flexible plastic makes this a risk.
- Start leaving sutures loose two sutures short of completing the anastomosis, then once the final stitch is in place remove the shunt by pulling gently on the tag: it should come free with minimal resistance.
- Put both ends of your suture under gentle tension and tie.

If a shunt cannot be placed, or the surgeon prefers not to use one, local blood flow can be temporarily reduced by applying bulldogs or deep stay sutures proximally and distally. The obvious disadvantage is that there is greater disruption in coronary blood flow than when a shunt is used.

## Blowers

Intermittent suction with a fine tip is useful when there is minimal blood flow. When blood flow into the operative field is constant a blowing

device provides a less obtrusive means of improving visibility. Blowing devices now blow humidified $CO_2$ which dissolves readily, reducing the risk of air embolism. Humidification prevents tissues drying out. The strength of the jet of $CO_2$ is usually controlled by adjusting the outlet valve on the gas cylinder. The normal setting is 4–8L/min. The amount of humidification can be controlled by adjusting the setting of the pressure bag, and by adjusting the inflow in the operative field. It is best to only direct the blower onto the tissues intermittently as prolonged exposure may result in macerating the tissues.

## Pitfalls of shunts and blowers

- It is possible to damage the vessel endothelium with repeated attempts to insert a shunt, which very occasionally may result in acute thrombosis of the vessel requiring revision of the anastomosis. If a smaller shunt will not easily enter the vessel, occlude the proximal vessel with a bulldog or a silastic loop.
- If the $CO_2$ is not humidified then the blower effectively desiccates the endothelium, again predisposing to early thrombosis.
- Always make sure you can see your needle tip relative to the shunt: if you suture the shunt inadvertently just carefully pull the suture back out and redo: if you only recognize this at completion of the anastomosis, it is usually possible to pull the shunt out enough to cut it free from the stray suture with a Potts scissors or an 11 blade.
- A shunt does not restore normal blood supply: it reduces the lumen that blood can flow through and occludes side branches, so ischemic changes can still result if anastomoses are not completed efficiently.

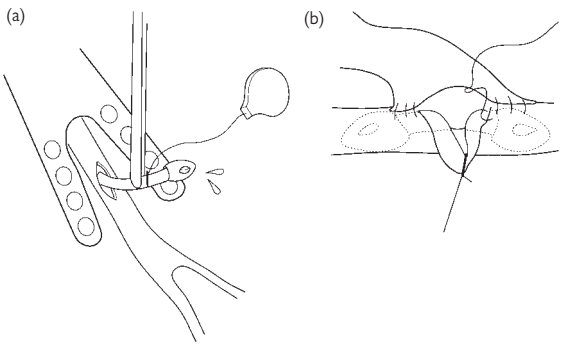



**Fig. 7.14** (a) Inserting the shunt. (b) Removing the shunt.

# Proximal anastomoses to aorta

## Key facts

- Although it is possible to perform the proximal anastomoses before the distal anastomoses, before or after institution of bypass, most surgeons perform them after completion of the distal anastomoses: this gives maximum flexibility to decide where the distal anastomosis is.
- They can be carried out with the cross-clamp still on, which reduces the number of times the aorta is side-clamped, reducing the risk of stroke, but prolonging the cross-clamp time (single-clamp technique).
- Alternatively the cross-clamp is removed after the last distal is performed and the proximals are carried out using a side-biting clamp.
- Using the Heartstring device allows a hand-sewn proximal anastomosis to be carried out without any clamp in place.
- Correct orientation and length is key to graft longevity.

## Technique

There are a variety of ways of performing this anastomosis. One method is described here. It is important to orientate the proximal anastomoses so that the conduit follows the most natural path, and is not kinked. It is also vital to measure the conduit accurately: if it is cut too long it is very likely to kink, and if it is cut too short both anastomoses will be under tension and thrombose early. Correct orientation is shown in Fig. 7.15. Taking wider bites of aorta than conduit around the heel, and wider bites of conduit than aorta at the toe produces the ideal 'cobra head' (Fig. 7.15).

- Look at the distal anastomosis while injecting blood or saline down the conduit, to establish the best way of orienting the conduit.
- Still injecting the conduit, after warning the perfusionist, place some occlusion on the venous line to fill the heart, to assess length.
- Divide it obliquely, keeping its orientation; if it is small conduit incise the heel 2–3mm, and fix it to the side-towels using a mosquito.
- Next, having palpated the aorta to avoid atheromatous plaques, prepare the aorta by dissecting the adventitia away from the area you plan to site your top-ends, using a McIndoe scissors.
- If you are using a side-biting clamp, ask the perfusionist to drop flows temporarily and apply the side-biter so that it is securely in place, and the toe of the clamp is flush with the aorta: too high and sutures catch on it, too low and the operative field will be full of blood.
- Make a 4mm stab incision with an 11 blade and create a round aortotomy using a 4mm hole punch, which must be kept clean of debris between uses to avoid introducing emboli into the ascending aorta.
- For the proximal anastomosis of circumflex vessels, the heel should lie at 2 o'clock (if 12 o'clock points towards the patient's head).
- Using double-ended 6/0 Prolene make the first stitch inside to out on the aorta at 12 o'clock, take the needle out of the needle-holder and secure it with a rubber-shod so that the maximum amount of suture is left on the other needle, and stitch forehand inside to out on the conduit, starting two stitches before the heel, at about 8 o'clock.

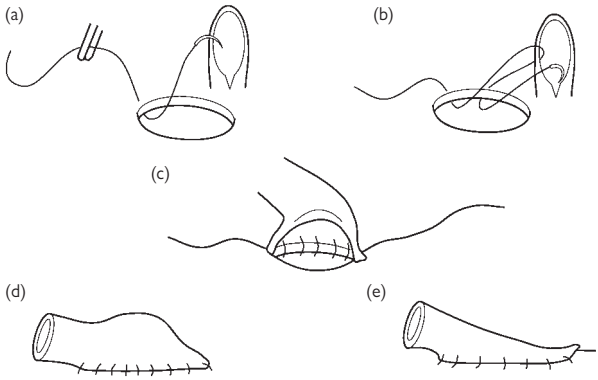



**Fig. 7.15** (a) First stitch. (b) Second stitch. (c) After parachuting down. (d) A satisfactory 'cobra head'. (e) An unsatisfactory one.

- Continue forehand outside to in on the aorta at 1 o'clock.
- Make even, full-thickness bites about 1mm from the edge of the conduit, 2mm from the edge of the aorta, and match the size of the conduit to the aortotomy so that a 'cobra-head' is produced.
- When you reach 6 o'clock on the aorta parachute the conduit down by pulling alternatively on the ends of the suture, and on the conduit to take up any loops: getting round to 6 o'clock makes the last stitches before you tie the anastomoses much easier, as they are not hidden by the vein, and can be done forehand.
- Check that there are no loops or crossed stitches.
- Change needles and suture the nearside forehand, from outside to in on the vein, and inside to out on the aorta in one pass, except for the first stitch which is better done in two passes.
- Tie down, and place a bulldog clamp across the conduit if there is not one already.
- For grafts to the right coronary system start at the toe of the conduit so that it lies at 12 o' clock, and for grafts to the LAD and diagonal vessels place the heel at 4 o'clock on the aorta (Fig. 7.15c).
- Remove the side-biting clamp if one is on, and prick the conduit to de-air it before removing the bulldog clamps: there should a full 'cobra head' rather than a flattened one, and no kinks or twists in the conduit.
- To repair any leaks from top ends, ask the perfusionist to 'go down on flow' and suture from conduit to aorta (backhand if necessary) usually as a figure-of-8.

# Redo coronary bypass surgery 1

Numbers of redo coronary artery bypass operations have decreased steadily from a peak of around 25–30% of all coronary surgery in the 1980s to <5% today. This is probably due to several factors:

- Advances in operative technique including the use of arterial grafts extending the patency rates of grafts: use of BIMA is associated with a lower rate of redo surgery than use of single IMA, which in turn has a lower rate of reoperation than no IMA.
- Better understanding of primary revascularization strategy leading to less need for reoperation to graft untreated native vessel lesions.
- Advances in secondary prevention including the now near universal use of aspirin, statins, and ACE inhibitors.
- Increased use of PCI to treat native and conduit stenosis.
- Better understanding of who benefits and who does not benefit, leading to narrowing of the indications for redo CABG.

## Etiology

In the 1960s and 1970s, the early period of CABG, the indications for redo surgery was early graft failure, which occurred in almost 1/3 of patients, and progression of atherosclerotic disease in non-grafted vessels, which occurred in >1/2 of patients. Improvements in operative technique and decision-making meant that by the 1980s, the commonest indications had changed: 92% of cases were for late graft failure, and reoperation for progression of atherosclerosis in non-grafted vessels was rare. Risk factors for redo CABG include:

- No use of IMA in original operation.
- Incomplete revascularization in original operation.
- Young age.
- Normal LV.
- 1VD or 2VD.
- Poor control of hypertension, hypercholesterolemia, smoking.

## Pathophysiology

The pathophysiology of ischemia is similar to that in unoperated patients, but the underlying pathology is highly variable (📖 p303). In the majority of patients presenting with ischemia within a year of coronary artery surgery, the grafts are patent: ischemia is attributable to incomplete revascularization. Late presentation is due to SVG atherosclerosis: patients with stenoses in conduits subtending the LAD do particularly poorly.

## Indications for reoperation

The mortality of surgery is increased in reoperation: the mortality in UK and US national registries for primary CABG is 1–2% compared to 5–8% for redo CABG. The higher mortality is due to increased rates of postoperative infarction and reflects:

- More advanced age.
- Poorer LV function and more severe distal and diffuse disease.
- More aortic atherosclerosis and peripheral vascular disease.
- Atherosclerotic vein grafts prone to embolize when handled.

- Risk presented by resternotomy.
- Risk of compromising patent graft to LAD territory in resternotomy.
- Difficulty in securing myocardial protection with patent LIMA to LAD.
- Less adequate conduit availability.
- Hypoperfusion if arterial conduit used to replace stenotic vein graft.

## ACC/AHA guidelines for redo CABG

***Class I***

- CABG should be performed in patients with prior CABG for disabling angina despite optimal non-surgical therapy (if angina atypical objective evidence of ischemia should be obtained. (*Level of evidence: B.*)
- Coronary bypass should be performed in patients with prior CABG without patent bypass grafts but with Class I indications for surgery for native vessel CAD (LMS, 3VD). (*Level of evidence: B.*)

***Class IIa***

- Coronary bypass is reasonable in patients with prior CABG and bypassable distal vessel(s) with a large area of threatened myocardium by non-invasive studies. (*Level of evidence: B.*)
- Coronary bypass is reasonable in patients who have prior CABG if atherosclerotic vein grafts with stenoses >50% supplying the LAD or large areas of myocardium are present. (*Level of evidence: B.*)

There are no randomized controlled trials of redo surgery versus PCI or medical intervention, all data is based on retrospective cohort analyses.

- There is little difference in 5-year survival between operated and non-operated patients with stenotic vein grafts.
- Late vein graft stenoses >50%, supplying either the LAD territory or large areas of myocardium are associated with a worse survival if managed medically.
- There is probably no survival benefit from reoperation in patients with patent LIMA to LAD grafts, and they should be offered PCI if suitable.
- Reoperated patients achieve better symptom control than medically treated patients.
- PCI has good results in early vein graft stenoses, native coronary disease, and anastomotic strictures, but is much less successful at treating late atherosclerotic vein graft stenosis.
- The 5-year survival after redo CABG is just over 75%, the 10-year survival is just under 50%.
- Emergency redo surgery has a high mortality.

## Summary of indications for redo CABG

1 Viable myocardium.
2 Supplied by stenotic vessels.
3 With suitable distal targets.

# Redo coronary bypass surgery 2

## Key facts

The surgeon must have a complete understanding of the native coronary anatomy, the bypass graft anatomy, and the relationship of major structures to the sternum, as well as a plan for conduit selection. This requires:

- Old coronary angiograms as well as one within 6 months.
- Previous operation notes, confirmed by clinical examination.
- Chest CT in all patients: CT angiography if renal function permits.

## Preoperative planning

***Patent LIMA to LAD anastomosis.***

Review the indications for surgery (📖 p343). This makes resternotomy potentially hazardous: if the LIMA is damaged before the heart is on bypass the result may be intractable VF, and has implications for myocardial protection (📖 p347).

***Right ventricle or atrium, aorta or grafts adherent to sternum***

This can only be reliably identified with a CT or MRI scan, which is the rationale for obtaining one in all reop patients to assist planning.

- *Very high risk:* e.g., aneurysmal aorta, patent LIMA or SVG adherent to midline of sternum: safest strategy may be to fully heparinize, cannulate peripherally, and go on bypass prior to sternal re-entry. Cooling gives the option of circulatory arrest in case of catastrophic bleeding but requires strategy for LV venting in case of fibrillation (📖 p346). Left and right lateral thoracotomy approaches, or mini-thoracotomy to aid dissection before sternotomy, are useful alternatives.
- *High risk:* e.g., right ventricle or dilated atrium adherent to sternum: peripheral arterial cannulation (5000U of heparin and tell perfusionist to flush cannula every 10min) with exposure of peripheral vein in readiness for cannulation if bypass required emergently is good option that allows mediastinal dissection without heparinization.
- *Moderate risk:* e.g., patent grafts but all well away from sternum on CT: consider dissecting out peripheral vessels prior to sternotomy, and ask to have coronary shunts in the room.

## Before induction of anesthesia

External defibrillator pads must be placed on the patient: you will not be able to use internal defibrillator paddles effectively until the heart is fully mobilized. Shave groins. If you are using axillary cannulation ensure the anesthesiologist sites a radial artery cannula in that arm (to detect hyperperfusion) *and* on the opposite site. Long cases need axillary or femoral A-lines as vasoconstriction often leads to damping of radial lines.

## Prepping and draping

Ensure that both groins, axillae, and legs are fully prepped and readily accessible. Mark all scars. Take the lines from the perfusionist.

## Incision

If the patient has a markedly hypertrophic scar, excise it. Otherwise make your incision in the line of the original scar whether or not it is precisely

in the midline. Making a fresh incision risks creating a patch of ischemic tissue between the two scars that will be slow to heal, and will leave two unsightly scars.

## Resternotomy

- Remove the sternal wires by pulling on them with the heavy needle holder and cutting one side of the loop so that the wire in its entirety is pulled out: alternatively leave them *in situ* and use them as a guard, preventing the sternal saw from lacerating the RV.
- The upper table of the sternum can be sutured with No. 5 Ethibond, then used as countertraction during sternotomy, and dissection afterwards: or a Lanes tissue holder may be applied to the xiphisternum.
- An oscillating saw is used for resternotomy, and the sternum must be lifted up, but not pulled laterally, to avoid tearing adherent RV.
- The safest technique is not to saw completely through the posterior table in the first pass: rotate a forceps handle to open up the anterior table and divide the posterior table carefully with a Mayo scissors.

## Dissection

It is difficult to predict how dense adhesions will be, but as a rule of thumb they become maximally dense up to 2–3 months after surgery, decreasing very gradually after that. Adhesions are most dense where there has been previous dissection, anastomosis, and cannulation sites. Diathermy and scissors are used. A 'no touch' technique is used around vein grafts to avoid dislodging atheroma and thrombus.

- Lifting up very gently on the left sternal edge, so as not to tear adherent RV or induce hemodynamic compromise, and staying close to bone, begin to free the mediastinal tissues from the chest wall using either cautery or a curved Mayo scissors.
- Place a sternal retractor once enough sternum is free and tension on all underlying structures has been relieved by dissection.
- Free the heart in the following order: diaphragmatic border (there is normally a plane of dissection here) left heart border, right heart border, aorta, right atrium.
- Cannulate and go on bypass, before continuing dissection of the apex, and finally the base of heart if necessary for grafting.
- Take great care to stay in the right plane, staying superficial to epicardial fat, avoiding 'skinning' the aorta (dissecting deep to the advential layer), and taking particular care round the thin-walled right atrium.

## Redo coronary surgery

Dealing with a patent LIMA to LAD is discussed on 📖 p347. Generally the original vein grafts are left alone, and handled minimally to avoid causing atheromatous emboli. The distals are performed in the usual manner: some surgeons anastomose to the stump of the original vein graft if there is no distal stenosis, and it is the vein graft itself which is occluded. The proximals are usually performed onto the hoods of the original vein grafts, as these are softer than the atheromatous aorta. There is a body of opinion that vein grafts should be replaced by vein grafts: one study reported a 20% incidence of hypoperfusion when LIMA was used to replace stenotic vein graft to the LAD.

# Problems specific to redo surgery

## Key facts

- Immediately life-threatening problems include aortic injury on sternal re-entry, and damage to patent grafts particularly the LIMA to LAD.
- Potentially very major problems include laceration of the right atrium or ventricle on sternal re-entry, subadventitial degloving of the aorta, and problems associated with peripheral cannulation (📖 p120).
- Potential problems requiring pre-planning are lack of conduit; failure to arrest the heart; lack of room on the aorta to cannulate, clamp plegia, vent and perform proximals; and lack of sites to graft distally.

## Injury to aorta on sternal re-entry

Spectacular bright red bleeding and rapid hypotension of you and patient.

***Management***

- Give heparin, tell anesthesiologist and perfusionist, get senior help.
- Likely impossible to effect a rapid primary repair as no access, because the sternum is still not fully divided and freed from mediastinum.
- Primary aim is to control bleeding, go on bypass and cool enough to stop circulation so you can dissect aorta free and repair it. Two options:
  1. Apply pressure to bring sternal edges together: if this seems to stop bleeding you have a short period of time to cannulate groin.
  2. If this doesn't work one salvage maneuver is to place large Foley catheter into where you think the hole is, blow up balloon with 5–10mL saline, and pull back gently which may tamponade bleeding without occluding ascending aorta. Cannulate peripherally and cool down.

***Avoiding this scenario***

- Always get and review the pre-op CT scan: choose re-entry and cannulation strategy based on proximity of aorta. Patients at very high risk of this injury should be heparinized, cannulated, and cooled in readiness for circulatory arrest, which makes this kind of injury much less likely and relatively easy to deal with if it does happen.
- Dividing the posterior table with the Mayo scissors rather than sternal saw (or using a thoracotomy/VATS to dissect the aorta free prior to sternal division) reduces the risk of this injury.

## Injury to patent LIMA or SVG

Notable for impressive bright red bleeding, ST segment changes, and VF.

***Management***

- Give heparin, tell anesthesiologist and perfusionist, get help.
- You may be able to effect a rapid primary repair, but commonly you cannot see to sew because of bleeding and adhesions.
- Usually possible to place a coronary shunt in the damaged conduit.
- Alternatively hook the cardioplegia line up to a catheter, place it pointing distally in conduit and ask the perfusionist to give continuous blood down it, while occluding any proximal bleeding segment with your finger or bulldog.
- Priority is then to go on bypass to minimize ischemic insult.

*Avoiding this scenario*

- Review all pre-op coronary angiography so that you can select re-entry and mediastinal dissection strategy appropriately (see earlier).
- There is almost no reason to dissect out the LIMA in a reop: the further you are away from it, the less likely you are to injure it.

## Major injury to right ventricle or right atrium

Impressive amounts of very dark blood, hypotension.

*Management*

- Straightforward if you are already heparinized and cannulated: go on bypass (you will need vacuum assist to prevent airlock and ensure adequate drainage) and complete enough dissection to effect a repair with multiple pledgetted horizontal mattress 4/0 Prolene sutures.
- If you are not cannulated, management is more challenging:
  - Stop retracting and hold chest together to prevent further tearing.
  - Give heparin.
  - Expose and cannulate femoral artery.
  - Place a pump sucker in the RV and go on "suction bypass".
  - Meanwhile cannulate the femoral vein and switch to fem–fem bypass, with vacuum assist, and proceed as previously described.

*Avoiding this scenario*

- Choose your re-entry strategy based on careful evaluation CT or MRI.
- In high-risk cases one option is to cannulate, but instead of going on bypass exsanguinate patient into the pump prior to resternotomy to decompress the RV, transfusing back as soon as dissection is complete.
- Always pull up not laterally on sternum until RV is free of adhesions.
- Take particular care when freeing the RA: it may be necessary to leave a small cuff of pericardium attached to the atrium.

## Problems with myocardial protection

- A patent LIMA is an asset, not a problem: there is no need to dissect this out and occlude it. Your options are:
  - Cardioplegic arrest required: cool the patient to 28°C and give intermittent retrograde and antegrade cardioplegia. The LIMA will run cold blood, protecting the heart and maintaining arrest.
  - Perform mitral and or coronary surgery with the heart beating or fibrillating. If there is even mild AI a large LV vent is needed.
- Retrograde cardioplegia largely avoids the need to figure out ways of giving cardioplegia down old vein grafts: give antegrade cardioplegia down new ones as you construct them.

## What to graft

- Many surgeons replace all vein grafts older than 5 years.
- Replace vein grafts with >50% stenosis feeding viable myocardium.
- Epicardial scarring may make target location difficult.
- Existing grafts provide a guide: blind dissection is sometimes needed.
- The principles are the same as first-time surgery with two exceptions:
  - An occluded vein graft can be replaced with an arterial graft.
  - A moderately stenosed vein graft should be replaced with a vein graft to avoid postoperative hypoperfusion, even in the LAD territory.

# Left ventricular aneurysm

## Morphology

The definition of a LV aneurysm and the difference between that and other types of LV scars is controversial. LV aneurysms are a result of MI: coronary disease is frequently confined to the LAD system. They are well demarcated, transmural, fibrous scars, devoid of muscle tissue, and much thinner than normal myocardium. 85% are located anterolaterally, near the apex of the heart. Only 5–10% are posterior, and these are mostly false aneurysms. Externally pericardium is often densely adherent. Internally, trabeculae are replaced by smooth fibrous tissue, and mural thrombus is often attached to the wall. The thrombus sometimes calcifies, as may the overlying pericardium.

## Structural and functional changes

LV aneurysms once occurred in up to 50% of survivors of large MIs, but this percentage is falling largely thanks to thrombolysis and angioplasty. Within 1 week of a large MI wall thinning and LV dilatation may be visible on echocardiography or ventriculography. Fibrous tissue takes at least 1 month to form. The LV wall is mostly composed of necrotic muscle tissue in the days after an MI, and this is when it is at greatest risk of rupture. Hyalinized fibrous tissue eventually replaces the muscle tissue. During systole the involved segments of wall are either immobile (akinetic) or bulge outwards (dyskinetic). If >20% of the LV wall is affected, non-aneurysmal portions of the LV also increase in size and thickness in response to the additional workload. True posterior aneurysms are associated with ischemic MR. RV function may be impaired as a result of a dyskinetic ventricular septum, pulmonary hypertension, and the tamponading effect of an enlarged LV.

- *Intractable VT* occurs most commonly in patients with poor LV, particularly where the ventricular septum is involved.
- *Postinfarction LV free wall rupture* occurs in about 10% of patients with acute infarction, and accounts for up to 20% of early deaths. It generally occurs between 1–7 days post infarction, and is usually a gradual process that begins with small endocardial tears, progressing to a haematoma which dissects through the necrotic myocardium resulting in sudden tamponade.
- *False LV aneurysm* occurs after postinfarction LV rupture when the pericardium is densely adherent to the epicardium. A slowly enlarging hemopericardium or false aneurysm eventually results, with a narrow mouth and a strong tendency to rupture, unlike true LV aneurysms.
- *Left ventricular remodelling* occurs after MI: even in the absence of scarring LV geometry changes and dilatation occurs, and is associated with progression of symptoms and poor prognosis. Surgical ventricular remodelling (SVR) is designed to address this and has been shown to improve ejection fraction and symptoms (📖 p495).

## Symptoms

Patients may have angina because of coronary artery disease, and classically complain of shortness of breath, persisting long after the MI. Up to 1/3 of patients have symptoms related to VT. Although most LV aneurysms contain thrombus, thromboembolic complications are rare.

- The size of the aneurysm and the functional characteristics of the remaining ventricle are major determinants of outcome.

### Indications for surgery

- Symptomatic aneurysms.
- False aneurysms.
- Repeated thromboembolic events.
- Post infarction LV rupture.
- Coronary artery disease that fulfils the criteria for surgery (📖 p296).

## Results of surgery

Early mortality after LV aneurysm repair, with or without CABG, is 5–7%, a substantial improvement on mortality rates of up to 20% a few decades ago. The improvement in survival is due to better myocardial protection, an emphasis on concomitant CABG, better protection against embolization, and treatment of VT. Following surgery: 1-year survival is 85%, 3-year survival is 75%, 5-year survival is 65%, although some centre report 5-year survivals as high as 88%.

## The STICH trial

- The Surgical Treatment for Ischemic Heart failure study (STICH) hypothesized SVR when combined with CABG would decrease death and hospitalization from CHF compared to CABG alone.
- Patients enrolled if they had CAD amenable to CABG and EF <35%.
- 499 randomized to medical therapy + CABG, 501 to medical therapy, CABG + SVR (1200 patients randomized to CAG vs. medical therapy to look at second hypothesis concerning CABG vs. medical treatment)
- Jones *et al.*[1] found that adding SVR to CABG did not improve mortality, symptoms, or hospitalization for CHF compared to CABG alone in these patients.
- Several problems with the methodology that impact its conclusions were pointed out in a response by Buckberg & Athanasuleas[2] which can be summed up by their premise that the study examined the wrong operation in the wrong patients by the wrong surgeons:
  - Wrong operation: in previous reports a 40% reduction in LV size is needed for benefit, STICH planned a minimum reduction of 30%, but final data shows a mean reduction of only 19% in LV volumes.
  - Wrong patients: initial inclusion criteria required LV volume documented >60mL/m$^2$ and >35% akinesia, which changed in the final study to 'documented LV anterior wall dysfunction' with only 38% of patients having echo measurement of LV volume, and only half of patients having documented LV akinesia or scar.
  - Wrong surgeons: study proposal required 10 cases per center and 5 per surgeon to participate with strict outcome criteria, which changed to only 5 per surgeon and looser outcome criteria.

## References

1. Jones RH *et al.* Coronary bypass surgery with or without surgical ventricular reconstruction. *NEJM* 2009; **360**:1705–17.
2. Buckberg GD & Athanasuleas CL. The STICH trial: Misguided conclusions. *J Thorac Cardiovasc Surg* 2009; **138**:1060–4.

# Left ventricular aneurysmectomy

## Key facts

- There are two main techniques: the classic technique is excision of the aneurysm and linear closure, and the *Dor procedure* involves patch closure or 'endoaneurysmorrhaphy', remodeling the ventricle (📖 p495).
- In both techniques the set-up is as described for CABG, including harvesting conduit if bypasses are planned (📖 p324–325). Particular care is taken to minimize handling the heart, in order to avoid dislodging mural thrombus.
- Myocardial protection is key particularly where LV function is compromised . Anterograde and retrograde cardioplegia is given. The heart is vented through a right superior pulmonary vein or root vent.

## Linear closure

- After opening the LV loose thrombus is removed.
- The incision is extended round the aneurysm leaving a rim of scar tissue.
- The LV is irrigated to clear any residual thrombus.
- The edges of the incision are brought together so that the LV is distorted as minimally as possible, and the incision is closed using Teflon strips to buttress interrupted horizontal 0/0 Ethibond mattress sutures, followed by a second layer of continuous sutures (Fig. 7.16).
- It is important to avoid suturing any portion of the LAD.
- Before completely closing off the incision, the heart is de-aired: the perfusionist occludes the venous line to fill the heart and the anesthesiologist inflates the lungs to evacuate air from the pulmonary veins air can be removed by aspirating the LV apex in the standard manner.
- The distal anastomoses are then performed as described on 📖 p328.
- This technique is used to treat post-infarction LV free wall rupture.

## Patch reconstruction

The aneurysm is incised but not excised. The recommended incision is in the thinnest portion of the aneurysm, parallel to the interventricular groove. A continuous purse-string (Fontan) suture using 2/0 Prolene is run along the line of demarcation of between the scar and the contractile myocardium (Fig. 7.16). A patch of PTFE which is sometimes backed with pericardium, measuring about 2 × 3cm, is then sewn into place, incorporating the purse-string suture and the heart is de-aired in the same way described above. The aneurysm is closed over the patch to reinforce it. The distal coronary anastomoses are then performed as described on 📖 p328. The LV cavity is often sized using a shaped balloon inflated with saline to a volume 50–60 × BSA.

## Mitral valve surgery

Mitral valve repair (papillary muscle repositioning with a sling around the papillary muscles, together with some form of annuloplasty) or replacement can be carried out via the ventriculotomy if indicated, although a standard left atriotomy approach after closure of the aneurysm approach is more familiar to most surgeons.

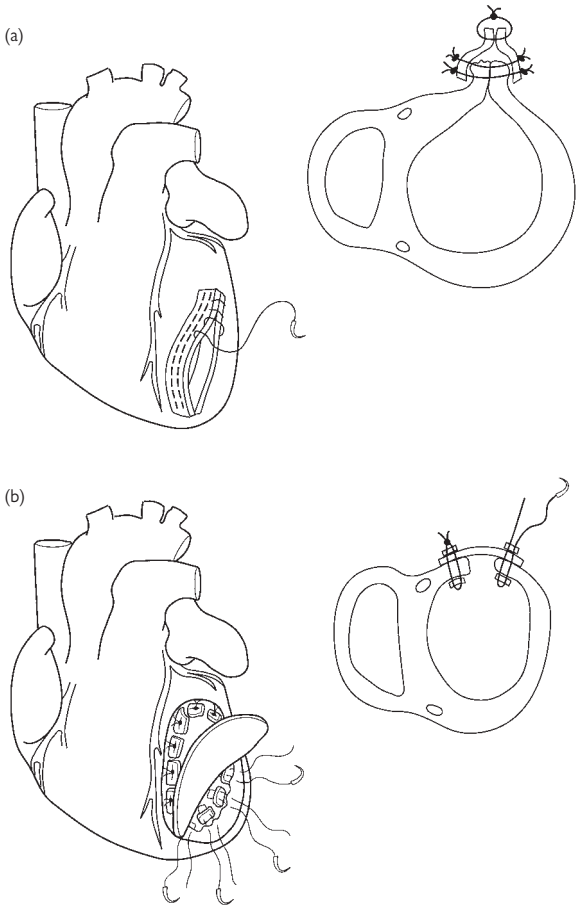



**Fig. 7.16** Aneurysmectomy. (a) Linear closure. (b) Patch closure.

## Postoperative management

Postoperative management is similar to CABG patients, with considerations for ventricular function which is likely to be particularly poor in the immediate postoperative period. Many patients will benefit from IABP, and may be on infusions of milrinone or dobutamine for several days after surgery. Patients with mural thrombus should be anticoagulated for at least 3 months after surgery.

# Ischemic ventricular septal defect

## Key facts

- Post-infarction ventricular septal defect (VSD) is a hole in the interventricular septum caused by rupture of acutely infarcted myocardium, in about 1–2% of patients with acute MI, classically 5 days post MI.
- Post-infarction VSD is an indication for urgent surgery (untreated survival is very poor), but operative mortality is around 30–40%.

## Morphology

- Ischemic VSDs are located in the anterior or apical area of the septum in 60% of cases, caused by infarcts in the LAD territory, and in the posterior part of the septum in 20–40% of patients as a result of inferior MI.
- Classically patients do not have a well-developed collateral coronary circulation, and the coronary obstruction is usually complete, rather than a high-grade stenosis. Important stenoses usually exist in the right coronary circulation, and concomitant RV infarction or severe ischemia with profoundly impaired RV function is an important feature of any ischemic VSDs.



Posterior VSDs are often accompanied by mitral valve regurgitation secondary to papillary muscle ischemia, infarction, or rupture.

## Structural and functional changes

The left-to-right shunt is normally large, with a pulmonary to systemic blood flow ratio (Qp/Qs) of >2.0. PAWP, LAP, LVEDP, and PAP are all raised. RV function is frequently profoundly impaired. Although this was once attributed to failure to adapt to the sudden increase in RV preload and afterload, the important role of RV ischemia and infarction described above is now recognized. About 40% of patients that survive ischemic VSD go on to develop LV aneurysms.

## Symptoms

The symptoms are hard to distinguish from that of the precipitating MI: the first sign of the ischemic VSD is usually a new pansystolic murmur, commonly at the lower left sternal edge, and frequently associated with the signs and radiographic evidence of new pulmonary venous congestion. The survival without surgery is extremely poor:



- 75% survival at 24h.
- 30% survival at 2 weeks.
- 10–20% survival at 4 weeks.

## Indications for surgery

Because of the poor survival, post-infarction VSD is almost invariably an indication for operation. The best results are for surgery performed after the acute phase in patients with a good cardiac output, no pulmonary hypertension or edema, and satisfactory renal function, but emergency surgery is indicated for the vast majority of patients who present with cardiogenic shock and severe pulmonary edema.

## Results of surgery

- Hospital mortality after VSD repair is about 40%.
- 5-year survival is about 50%.
- 10-year survival is about 30%.

# Ischemic ventricular septal defect repair

## Key facts

Preoperative management and assessment is critical: these patients are usually extremely unwell. If a PA catheter and IABP have not already been inserted, this should be done immediately. If stable enough the patient should undergo coronary angiography prior to transfer to the OR.

## Set-up

- Open the pericardium to one side, and harvest a patch for later use.
- Bicaval cannulation with arterial return to the ascending aorta. Caval tapes: the patient is cooled to 30°C.
- Retrograde cardioplegia is often used as an adjunct to antegrade cardioplegia administered via the aortic root.
- The LIMA is not normally used. Rapid institution of full cardiopulmonary bypass followed by speedy revascularization is the priority: saphenous vein graft can be taken much more quickly without delaying bypass, and while the VSD repair is effected.

## Technique

There are two main techniques (primary repair and infarct exclusion) that are used, depending on the area infarcted. Primary repair is the most popular technique. The septum is usually approached via the LV.

*Anterior VSD*: the incision is made in the infarcted surface of the antero-lateral LV. The defect in the septum is usually immediately beneath the incision (Fig. 7.17a). The defect is not resected. A patch repair with PTFE, autologous or bovine pericardium is carried out so that the whole of the LV aspect of the septum is covered (Fig 7.17c). Infarct exclusion excludes the LV cavity from the infarcted myocardium by suturing a larger patch so that it extends all the way to the non-infarcted endocardium of the antero-olateral LV wall. Pledgetted horizontal mattress sutures of 2/0 Prolene are placed so that the pledgets are on the RV side of the repair, and sutures occupy the non-infarcted myocardium. Sutures must be placed quite close together. If it is possible to close the incision in the LV wall primarily this is done with interrupted horizontal mattress sutures of 0/0 Prolene placed across four layers of pericardium or PTFE felt trips (Fig. 7.17), and reinforced with continuous 2/0 Prolene over and over stitch. Otherwise a patch closure should be carried out.

*Apical VSD:* apical VSDs are usually associated with apical infarction: the apex of the heart is amputated if the VSD is immediately adjacent to the apical infarction (Fig. 7.17b). The defect, and effectively the VSD is closed primarily using interrupted horizontal mattress sutures of 0/0 Prolene placed across four layers of pericardium or PTFE felt trips (Fig. 7.17b), and reinforced with continuous 2/0 Prolene over and over stitch.

*Posterior VSD*: these are more difficult to repair. The apex of the heart is elevated with stay sutures, or a suction cup. A vertical incision is made

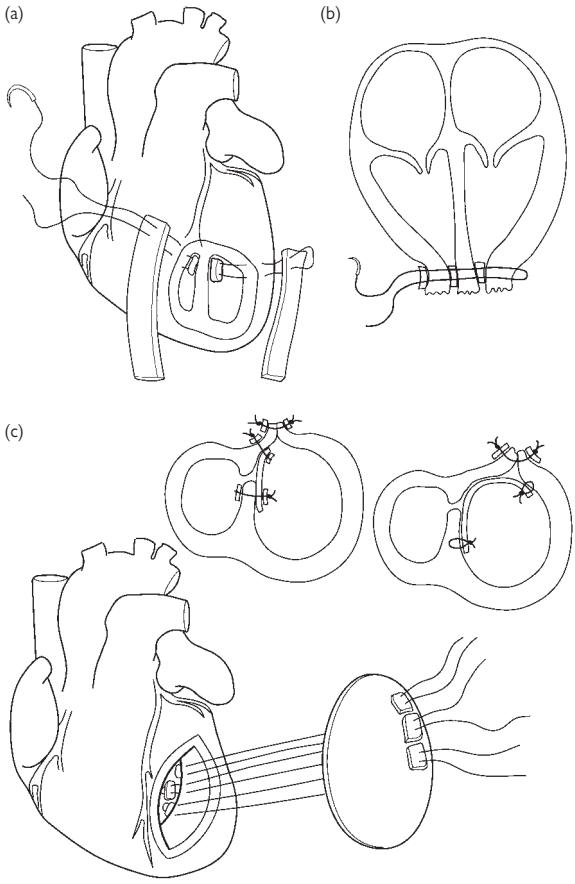



**Fig. 7.17** (a) Anterior VSD repair. (b) Apical VSD repair. (c) Posterior VSD repair.

in the infarcted posterior wall of the LV, parallel to the PDA. The VSD is resected and a patch attached to the LV aspect of the remaining non-infarcted septum as described *previously* (Fig. 7.17a). Depending on how much of the free RV and LV wall is infarcted, the options for closure are patch closure of the RV, primary closure of the LV buttressed by an external patch, or infarct exclusion.

# Ischemic mitral regurgitation

## Key facts

- Ischemic MR is MR caused by ischemic heart disease, rather than MR co-existent with ischemic heart disease: the management and outcomes of ischemic and non-ischemic MR are very different.
- Chronic ischemic MR is usually Carpentier type IIIb (📖 p400) due to *posterior leaflet restriction* from displacement of posteriomedial papillary muscle due to inferior LV ischemia. Anterior ischemia tends to result in symmetrical annular dilatation (type I MR). Both should be repaired with rigid, down-sized complete rings.
- Acute, new MR following an MI may be type II (prolapse) due to *papillary muscle rupture*. This usually requires replacement rather than repair, and is associated with high operative mortality.

## Causes of ischemic MR

- Annular dilatation (type I).
- Papillary muscle rupture (type II)—acute.
- Papillary muscle necrosis (type II).
- Changes in configuration of the LV (type IIIb).
- Global and segmental LV dysfunction (type IIIb).

## Morphology

The morphology of acute ischemic MR differs from that of chronic MR.

*Acute ischemic MR:* MR is a common complication of acute MI. Moderate to severe MR is present in up to 20% of patients in the first few days after an MI. It is a predictor of mortality. Acute ischemic MR is rarely caused by papillary muscle dysfunction: it is much more commonly due to LV systolic dysfunction preventing full leaflet coaptation. When papillary muscle rupture does occur, it may be partial or complete resulting in flailing of both anterior and posterior leaflets. In 75% of cases the posteromedial papillary muscle is ruptured, and in 25% of cases the anterolateral. The difference is probably due to the dual supply of the anterolateral papillary muscle (LAD and Cx *vs.* single supply of the posteromedial PM (PDA or Cx) (📖 p46).

*Chronic ischemic MR:* this is type IIIb (restriction of normal leaflets) caused most commonly by papillary muscle displacement due to LV dysfunction and/or dilatation, or type I MR due to annular dilatation. Type IIIb is localized to the P3 area, due to the more common involvement of the supporting posteromedial papillary muscle in infarction.

## Structural and functional changes

Papillary muscle rupture presents as an acute event, resulting in acute cardiogenic shock associated with raised LAPs, raised RVEDP, and pulmonary edema. The pathophysiology of acute MR is described on 📖 p401. The course of chronic ischemic MR is similar to that described on 📖 p400, but the degree of MR varies depending on LV function and preload.

## Symptoms

Acute ischemic MR presents a few hours to weeks after an MI with pulmonary edema, hypotension, and an apical pansystolic murmur which is frequently absent in complete papillary muscle rupture. Chronic ischemic MR presents in much the same way as MR of other etiology (📖 p401). The survival of papillary muscle rupture without surgery is poor:

- 25% of patients survive 24h after complete papillary muscle rupture.
- 70% of patients survive 24h after partial papillary muscle rupture.
- MR after MI is a strong predictor of mortality.

## Surgery for ischemic MR

### *Indications for surgery*

In chronic ischemic mild MR, full revascularization may be sufficient to improve valve function. Moderate to severe chronic ischemic MR should be treated with mitral repair at the time of CABG. Patients with severe LV dysfunction and LVEDD >7.5cm gain little clinical benefit from conventional surgery and should be considered for medical management, ventricular assist device, or transplant (📖 p494). Acute severe MR is an indication for urgent surgery.

### *Techniques of ischemic mitral surgery*

The principles of mitral valve surgery are described on 📖 pp404–410. The aspects specific to the ischemic mitral valve are:

- Acutely unwell patients require preoperative insertion of an IABP.
- LIMA harvest may not be appropriate in the unstable patient.
- Retrograde cardioplegia is a useful adjunct in acute coronary occlusion.
- The distal anastomoses are performed before the mitral valve prosthesis is inserted to avoid lifting the heart with a rigid structure sutured to the annulus, which can cause AV dehiscence.
- The proximals can be performed before or after mitral surgery.
- MVR is indicated in total papillary muscle rupture: particular care must be taken with the ring sutures as the tissues are friable.
- Mitral annuloplasty with a rigid down-sized ring is preferable for type IIIb and type I MR (restricted and annular dilation), with secondary chordal cutting to P2 and P3 usually required to achieve a good surface of coaptation and a durable repair.

### *Outcomes of surgery*

- Surgery for acute ischemic MR has a hospital mortality of up to 70%.
- Repair of chronic ischemic MR has a hospital mortality is 3–10% depending on comorbidity. Poor LV function, extensive coronary artery disease, previous anterior infarction, and advanced age are strong predictors of mortality.
- Multiple retrospective studies show variable benefit of mitral repair over replacement for ischemic MR, but the repair strategies were flawed (few surgeons were consistently using down-sized rigid annuloplasty rings or employing chordal cutting, and rates of residual and recurrent MR were relatively high). The NIH is currently conducting multicenter RCTs of repair vs. replace in severe ischemic MR, although with an echocardiographic rather than a clinical primary endpoint.

# Risk of coronary surgery

## Crude survival

Risk-adjusted outcome is a cornerstone of understanding the results of surgery, planning surgery and predicting operative mortality for individual patients. Without adjusting for risk, after isolated CABG:

- 30-day survival is 98%.
- 1-year survival is 97%.
- 5-year survival is 92%.
- 10-year survival is 81%.
- 15-year survival is 66%.

## STS score and EuroSCORE

These are both weighted, additive scoring systems that give scores for a number of patient-related and surgical variables, that are added together to give an indication of an individual patient's risk of perioperative mortality. The EuroSCORE (inside front cover) is derived from analysis of a pan-European database of surgical patients. It improves on the earlier Parsonnet score, which was based on a North American patient database. The EuroSCORE produces a number that equals the predicted percentage mortality but tends to overestimate risk. The STS score is a little more complex but provides quite accurate risk prediction.

## Hospital mortality

The variables most predictive of early hospital mortality are listed in Box 7.2. Other variables associated with 30-day mortality include recent STEMI, diabetes, end stage renal disease, valvular disease, and COPD.

**Box 7.2 Key predictors of earlier hospital mortality**

- Older age.
- Female gender.
- Previous CABG.
- Urgency of operation.
- Increasing LV dysfunction.
- LMS disease.
- Increasing extent of CAD.

## Risk factors for death after isolated CABG

- Patient risk factors:
  - ↑ or ↓age.
  - ↑weight or ↑weight/height ratio.
- Symptoms of reversible ischemia:
  - CCSA class.
  - No angina.
  - Unstable angina.
- Cardiac comorbidity:
  - Mitral regurgitation.
  - Aortic regurgitation.
  - Chronic atrial fibrillation.
  - Ventricular tachycardia/fibrillation.
  - Cardiac pacemaker.
- Non-cardiac comorbidity:
  - BMI >30 or <20.
  - History of vascular disease.
  - History of cerebrovascular disease.
  - Previous vascular surgery (non-carotid).
  - History of smoking.
  - ↓$FEV_1$.
  - History of renal failure.
  - Renal dialysis.
  - Serum creatinine.
  - Hypertensive.
  - ↑grade of diabetes and age.
  - History of malignancy.
  - History of hepatic disease.
  - ↑triglyceride level.
- Left ventricular function:
  - Cardiogenic shock.
  - Clinical status.
  - ↓EF.
  - Date of operation.
  - Limitation by heart failure.
- Coronary disease:
  - 90% LMS.
  - ↑number of diseased systems.
  - 3VD.

# Results of coronary surgery

## Morbidity and survival

Risk factors for reduced survival after CABG are shown in Table 7.3. Postoperative complications are described on 📖 pp253–290. The most significant complications of CABG are:

- Significant cerebrovascular event in 1–2%.
- Deep sternal wound infection in ~1–2%.
- Renal dysfunction (7%) requiring dialysis (1%).
- Resternotomy for bleeding 1–2%.
- Major GI bleed requiring EGD or surgery 1–2%.

## Recurrent angina

Angina recurs in 40% of patients by 10 years postoperatively. The highest rates of recurrence are at 3 months postoperatively, related to incomplete revascularization and early graft occlusion, and 3 years postoperatively as a result of progression of coronary disease and graft stenosis. Patients who experience recurrent angina, however, have a similar survival to patients that remain symptom free. Use of the LIMA as conduit has a much better graft patency rate in the long term, which is not, surprisingly, very correlated to improved freedom from symptoms.

## Myocardial infarction

Incidence of perioperative MI is about 2–5% depending on definition. Postoperative MI is a predictor of mortality.

## Quality of life

There are a number of quality of life measures, but the most important factors are; (1) freedom from angina, (2) freedom from medication, hospitalization, and reintervention, and (3) preserved exercise capacity. Most surviving patients have a satisfactory quality of life which begins to decline 5 years after surgery, mirrored by the increase in recurrent angina. Returning to work is very dependent on a number of factors. <20% of patients with an older age at operation, lower educational standard, longer absence from work, lower income and more physical job preoperatively, longer and greater angina, peripheral vascular disease, greater alcohol intake, emergency operation, and who fail to attend a cardiac rehabilitation programme return to work. 80% of patients without those factors return to work postoperatively. The amount of medication taken by patients is usually lower postoperatively. Functional exercise capacity is generally improved more by CABG than by medical treatment for up to 10 years. The improvement depends on preoperative LV function, graft patency and completeness of revascularization, and is related to the maximal heart rate achievable during exercise.

## Graft patency

The patency rates of various conduit are summarized on 📖 p302.

**Table 7.3** Incremental risk factors for death after primary CABG

| | Impact on survival | | |
|---|---|---|---|
| **Risk factors** | **Early** | **Constant** | **Late** |
| **Coronary operations:** | | | |
| Greater disease complexity (higher SYNTAX score) | | | • |
| Non-use of LIMA | | • | • |
| Use of patch grafts | | | • |
| ↑Ratio of distals to conduits | • | | • |
| ↑Coronary endarterectomies | • | | |
| ↑Proportion of distals to small vessels | • | | • |
| Incomplete revascularization | • | | • |
| **Concomitant procedures:** | | | |
| Left ventricular incision or plication | • | | |
| **Institutional experience:** | | | |
| Low surgeon volume | • | | |

## Reintervention

Non-risk-adjusted freedom from re-intervention after CABG is around 90–95% at 3 years (based on SYNTAX 📖 p295). Vein graft atherosclerosis is the commonest cause of reintervention, followed by progression of native vessel disease. More frequent use of LIMA to LAD anastomoses has reduced the frequency of reoperation and lengthened the interval between the first and the second coronary operation. The other main risk factor for reoperation is younger age.

## Postoperative progression of native vessel disease

High-grade stenoses in native vessels tend to progress whether they are proximal or distal to the anastomoses. Low-grade stenoses distal to the anastomoses do not tend to progress, although there is some evidence that suggests the opposite is true if the graft is non-functioning. 25–50% of lesser stenosis in non-grafted arteries progress within 5 years. In vessels that were disease-free at the time of re-operation new lesions will be detected in 15% at 5 years.

## Left ventricular function

Segments of the LV that were dyskinetic or akinetic preoperatively usually improve, even in areas of scarring post MI. This supports hibernation theory (📖 p75). The improvement is reflected in regional perfusion defects which are improved after CABG in >65% of patients. Where preoperative ejection fraction recovery is limited but still usual, failure to improve LV function normally reflects incomplete revascularization. Coronary flow reserve returns to normal, reflected by a return of the normal increase in ejection fraction in response to exercise.

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